NACCA COURSES • MATHEMATICS

MATHS QUEST

BASIC 5

Explore 54 adventures in number, algebra, geometry, measurement and data.

54	66	810
LESSONS	CURRICULUM INDICATORS	ORIGINAL PRACTICE ITEMS

PRINTABLE LEARNER'S GUIDE · 2026 LEARNING EDITION

Your Maths Quest Starts Here

This book helps you think like a mathematician. Explore an example, try five guided Play tasks, then solve ten different Challenge problems.

LOOK & EXPLORE

Read the lesson goal, investigate its diagram and follow a worked example.

PLAY · 5 QUESTIONS

Use what you have just learned. Check your methods with a friend or adult.

CHALLENGE · 10 QUESTIONS

Try new numbers and fresh situations. Show your method, not only your answer.

REFLECT

Explain why your method works. Use the answer keys for self-checking after you try.

SAFETY AND SMART PRACTICE

Use clean counters, paper, pencils and a ruler. Ask an adult before going outside to measure, observe traffic, or handle tools. All activities can be done safely with drawings.

Find Your Next Adventure

01	Modelling numbers to one million	Page 6
02	Reading and writing large numbers	Page 9
03	Finding numbers on a number chart	Page 12
04	Comparing and ordering whole numbers	Page 15
05	Rounding large whole numbers	Page 18
06	Skip counting forwards and backwards	Page 21
07	Roman numeral symbols up to C	Page 24
08	Converting Roman and Arabic numerals	Page 27
09	Finding factors up to 100	Page 30
10	Prime and composite numbers	Page 33
11	Even and odd numbers	Page 36
12	Highest common factors	Page 39
13	Multiples and factor relationships	Page 42
14	Mental strategies for multiplication facts	Page 45
15	Efficient mental multiplication	Page 48
16	Multiplying by two-digit numbers	Page 51
17	Division facts to 81	Page 54
18	Dividing three-digit numbers	Page 57
19	Multi-step word problems	Page 60
20	Adding and subtracting integers	Page 63
21	Finding equivalent fractions	Page 66

Find Your Next Adventure

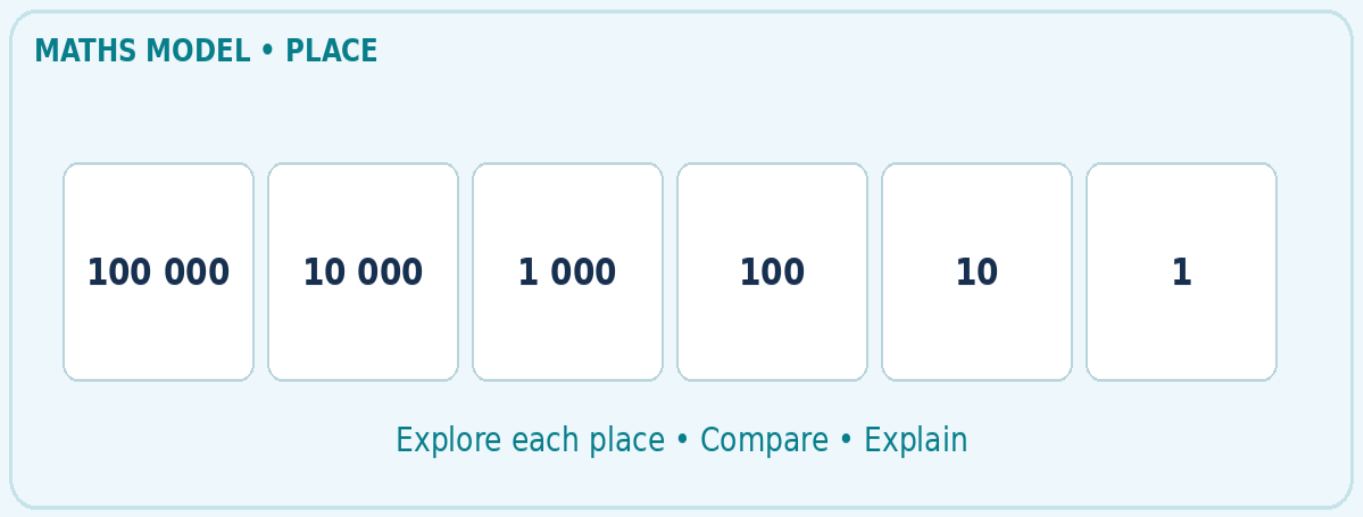
22	Comparing and ordering fractions	Page 69
23	Making common denominators	Page 72
24	Adding and subtracting fractions	Page 75
25	Multiplying a whole number by a fraction	Page 78
26	Finding a fraction of a number	Page 81
27	Modelling tenths, hundredths and thousandths	Page 84
28	Comparing decimals and fractions	Page 87
29	Rounding decimals	Page 90
30	Adding and subtracting decimals	Page 93
31	Multiplying with decimals	Page 96
32	Percentages of quantities	Page 99
33	Benchmark percentage facts	Page 102
34	Percentages in everyday situations	Page 105
35	Extending and describing patterns	Page 108
36	Predicting and drawing patterns	Page 111
37	Using patterns to solve problems	Page 114
38	Writing a pattern rule	Page 117
39	Building algebraic expressions	Page 120
40	Representing and solving equations	Page 123
41	Creating stories for equations	Page 126
42	Properties of squares and rectangles	Page 129

Find Your Next Adventure

43	Regular and irregular polygons	Page 132
44	Estimating and measuring perimeter	Page 135
45	Finding areas of rectangles	Page 138
46	Volume units and cubic blocks	Page 141
47	Different boxes with the same volume	Page 144
48	Comparing and measuring angles	Page 147
49	The four compass directions	Page 150
50	Reflections and mirror images	Page 153
51	First-hand and second-hand data	Page 156
52	Drawing and reading double-bar graphs	Page 159
53	Impossible, possible and certain	Page 162
54	Designing probability experiments	Page 165

Modelling numbers to one million

LEARNING GOAL Modelling numbers to one million



LEARN THE IDEA

A digit has a different value in each place: hundreds, thousands, ten thousands, hundred thousands and millions. multiply the digit by its place value.

WORKED EXAMPLE 1

In 524,668, what value does the digit in the 100s column represent?

Answer: 600

6 × 100 = 600.

WORKED EXAMPLE 2

In 415,318, what value does the digit in the 10s column represent?

Answer: 10

1 × 10 = 10.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Modelling numbers to one million

PLAY · LEARN THE STEPS

1. **In 174,769, what value does the digit in the 100,000s column represent?**
   A) 100,000 B) 1 C) 1,000,000 D) 10,000
2. **In 583,817, what value does the digit in the 10,000s column represent?**
   A) 800,000 B) 8 C) 8,000 D) 80,000
3. **In 529,071, what value does the digit in the 1,000s column represent?**
   A) 900 B) 90,000 C) 9 D) 9,000
4. **In 521,922, what value does the digit in the 1,000s column represent?**
   A) 10,000 B) 1 C) 1,000 D) 100
5. **In 134,524, what value does the digit in the 100,000s column represent?**
   A) 1 B) 10,000 C) 100,000 D) 1,000,000

CHALLENGE · DIFFERENT EXAMPLES

1. **A pupil at a market stall removes the contribution of the 1,000s digit from 938,988. What number remains?**
   A) 938,980 B) 930,988 C) 938,988 D) 931,988
2. **A record during a library project shows 780,719. What is the value of the digit in the 100s column?**
   A) 7,000 B) 7 C) 70 D) 700
3. **A six-digit number on a football field has 789,596 on a display. Which digit represents the 1,000s place?**
   A) 2 B) 9 C) 0 D) 4
4. **A pupil in a recycling club removes the contribution of the 10s digit from 324,127. What number remains?**
   A) 324,125 B) 324,127 C) 324,117 D) 324,107
5. **A record at a science exhibition shows 865,213. What is the value of the digit in the 10,000s column?**
   A) 6 B) 600,000 C) 60,000 D) 6,000

Modelling numbers to one million

CHALLENGE · APPLY WHAT YOU KNOW

6. **A record during a reading competition shows 301,592. What is the value of the digit in the 1,000s column?**
   A) 1,000 B) 100 C) 10,000 D) 1

7. **A six-digit number for a school fundraiser has 454,493 on a display. Which digit represents the 1,000s place?**
   A) 5 B) 4 C) 9 D) 7

8. **A pupil at a farm removes the contribution of the 100s digit from 467,965. What number remains?**
   A) 467,965 B) 467,165 C) 467,956 D) 467,065

9. **A record during a sports day shows 695,663. What is the value of the digit in the 10,000s column?**
   A) 90,000 B) 9 C) 900,000 D) 9,000

10. **A six-digit number in a craft workshop has 490,418 on a display. Which digit represents the 100,000s place?**
    A) 4 B) 9 C) 7 D) 5

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 100,000	2. 80,000	3. 9,000
4. 1,000	5. 100,000	6. 930,988
7. 700	8. 9	9. 324,107
10. 60,000	11. 1,000	12. 4
13. 467,065	14. 90,000	15. 4

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Reading and writing large numbers

LEARNING GOAL Reading and writing large numbers

MATHS MODEL • WORDS

12 340	25 600	43 125	65 700	89 020

Explore each place • Compare • Explain

LEARN THE IDEA

Read large numbers in groups of three digits. say the thousands group first, then the hundreds, tens and ones.

WORKED EXAMPLE 1

Choose the number name for 404,030.

Answer: four hundred and four thousand thirty

404,030 is read as four hundred and four thousand thirty.

WORKED EXAMPLE 2

What number is written as “five hundred and twenty-seven thousand thirty”?

Answer: 527,030

Convert the place-value words to 527,030.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Reading and writing large numbers

PLAY · LEARN THE STEPS

1. **Choose the number name for 909,010.**
   A) nine hundred and eight thousand ten B) nine hundred and nine thousand one hundred and ten C) nine hundred and nine thousand ten D) nine hundred and ten thousand ten

2. **What number is written as “nine hundred and eighty-five thousand six hundred and seventy”?**
   A) 985,770 B) 985,670 C) 984,670 D) 986,670

3. **Choose the number name for 728,280.**
   A) seven hundred and twenty-eight thousand two hundred and eighty B) seven hundred and twenty-eight thousand three hundred and eighty C) seven hundred and twenty-seven thousand two hundred and eighty D) seven hundred and twenty-nine thousand two hundred and eighty

4. **What number is written as “eight hundred and thirty-nine thousand seven hundred and seventy”?**
   A) 840,770 B) 838,770 C) 839,870 D) 839,770

5. **Choose the number name for 350,340.**
   A) three hundred and fifty thousand four hundred and forty B) three hundred and fifty thousand three hundred and forty C) three hundred and forty-nine thousand three hundred and forty D) three hundred and fifty-one thousand three hundred and forty

CHALLENGE · DIFFERENT EXAMPLES

1. **A sign during a library project says “four hundred and eighty-seven thousand fifty”. Which numeral should be printed?**
   A) 488,050 B) 487,050 C) 486,050 D) 487,150

2. **A presenter on a football field reads the number 610,010. Which wording is accurate?**
   A) six hundred and ten thousand one hundred and ten B) six hundred and nine thousand ten C) six hundred and eleven thousand ten D) six hundred and ten thousand ten

3. **A sign in a recycling club says “two hundred and eighteen thousand nine hundred and seventy”. Which numeral should be printed?**
   A) 218,970 B) 219,970 C) 219,070 D) 217,970

4. **A presenter at a science exhibition reads the number 348,110. Which wording is accurate?**
   A) three hundred and forty-eight thousand one hundred and ten B) three hundred and forty-nine thousand one hundred and ten C) three hundred and forty-seven thousand one hundred and ten D) three hundred and forty-eight thousand two hundred and ten

5. **A sign during a reading competition says “four hundred and ninety-nine thousand seven hundred and seventy”. Which numeral should be printed?**
   A) 498,770 B) 499,870 C) 499,770 D) 500,770

Reading and writing large numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **A presenter for a school fundraiser reads the number 684,610. Which wording is accurate?**

A) six hundred and eighty-four thousand seven hundred and ten B) six hundred and eighty-four thousand six hundred and ten C) six hundred and eighty-three thousand six hundred and ten D) six hundred and eighty-five thousand six hundred and ten

7 **A sign at a farm says “fifty thousand three hundred and sixty”. Which numeral should be printed?**

A) 50,460 B) 51,360 C) 50,360 D) 49,360

8 **A presenter during a sports day reads the number 388,210. Which wording is accurate?**

A) three hundred and eighty-eight thousand two hundred and ten B) three hundred and eighty-nine thousand two hundred and ten C) three hundred and eighty-eight thousand three hundred and ten D) three hundred and eighty-seven thousand two hundred and ten

9 **A sign in a craft workshop says “eight hundred and forty-nine thousand four hundred and thirty”. Which numeral should be printed?**

A) 849,530 B) 850,430 C) 849,430 D) 848,430

10 **A presenter at a school fair reads the number 697,850. Which wording is accurate?**

A) six hundred and ninety-seven thousand eight hundred and fifty B) six hundred and ninety-eight thousand eight hundred and fifty C) six hundred and ninety-seven thousand nine hundred and fifty D) six hundred and ninety-six thousand eight hundred and fifty

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. nine hundred and nine thousand ten	2. 985,670	3. seven hundred and twenty-eight thousand two hundred and eighty
4. 839,770	5. three hundred and fifty thousand three hundred and forty	6. 487,050
7. six hundred and ten thousand ten	8. 218,970	9. three hundred and forty-eight thousand one hundred and ten
10. 499,770	11. six hundred and eighty-four thousand six hundred and ten	12. 50,360
13. three hundred and eighty-eight thousand two hundred and ten	14. 849,430	15. six hundred and ninety-seven thousand eight hundred and fifty

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding numbers on a number chart

LEARNING GOAL Finding numbers on a number chart

MATHS MODEL • CHART

25 000	25 500	26 000	26 500	27 000

Explore each place • Compare • Explain

LEARN THE IDEA

On a number chart, check the size of a jump. moving right or forward adds; moving left or backward subtracts.

WORKED EXAMPLE 1

What is 10 less than 58,250?

Answer: 58,240

58,250 − 10 = 58,240.

WORKED EXAMPLE 2

What is 1 more than 11,733?

Answer: 11,734

11,733 + 1 = 11,734.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Finding numbers on a number chart

PLAY · LEARN THE STEPS

1. **What is 1 more than 9,955?**
   A) 9,954 B) 9,965 C) 9,956 D) 9,957
2. **What is 100 more than 43,034?**
   A) 42,934 B) 43,134 C) 43,234 D) 44,034
3. **What is 10 more than 26,288?**
   A) 26,278 B) 26,308 C) 26,388 D) 26,298
4. **What is 100 less than 41,658?**
   A) 41,758 B) 41,458 C) 41,558 D) 40,658
5. **What is 1 less than 43,380?**
   A) 43,370 B) 43,378 C) 43,381 D) 43,379

CHALLENGE · DIFFERENT EXAMPLES

1. **A chart on a football field lists steps of 1. The highlighted box is 49,827; what number is immediately after it?**
   A) 49,829 B) 49,828 C) 49,837 D) 49,826
2. **A chart in a recycling club lists steps of 10. The highlighted box is 25,776; what number is immediately after it?**
   A) 25,766 B) 25,796 C) 25,876 D) 25,786
3. **A chart at a science exhibition lists steps of 100. The highlighted box is 34,209; what number is immediately after it?**
   A) 34,409 B) 34,309 C) 34,109 D) 35,209
4. **A chart during a reading competition lists steps of 100. The highlighted box is 18,807; what number is immediately before it?**
   A) 18,607 B) 17,807 C) 18,907 D) 18,707
5. **A chart for a school fundraiser lists steps of 100. The highlighted box is 72,651; what number is immediately after it?**
   A) 73,651 B) 72,851 C) 72,551 D) 72,751

Finding numbers on a number chart

CHALLENGE · APPLY WHAT YOU KNOW

6 **A chart at a farm lists steps of 1000. The highlighted box is 54,645; what number is immediately after it?**

A) 55,645 B) 64,645 C) 53,645 D) 56,645

7 **A chart during a sports day lists steps of 1000. The highlighted box is 62,413; what number is immediately before it?**

A) 63,413 B) 60,413 C) 52,413 D) 61,413

8 **A chart in a craft workshop lists steps of 1. The highlighted box is 37,378; what number is immediately before it?**

A) 37,379 B) 37,376 C) 37,377 D) 37,368

9 **A chart at a school fair lists steps of 1000. The highlighted box is 44,262; what number is immediately after it?**

A) 45,262 B) 46,262 C) 54,262 D) 43,262

10 **A chart in the class garden lists steps of 1000. The highlighted box is 26,326; what number is immediately before it?**

A) 16,326 B) 24,326 C) 25,326 D) 27,326

ANSWER CHECK · LOOK ONLY AFTER TRYING

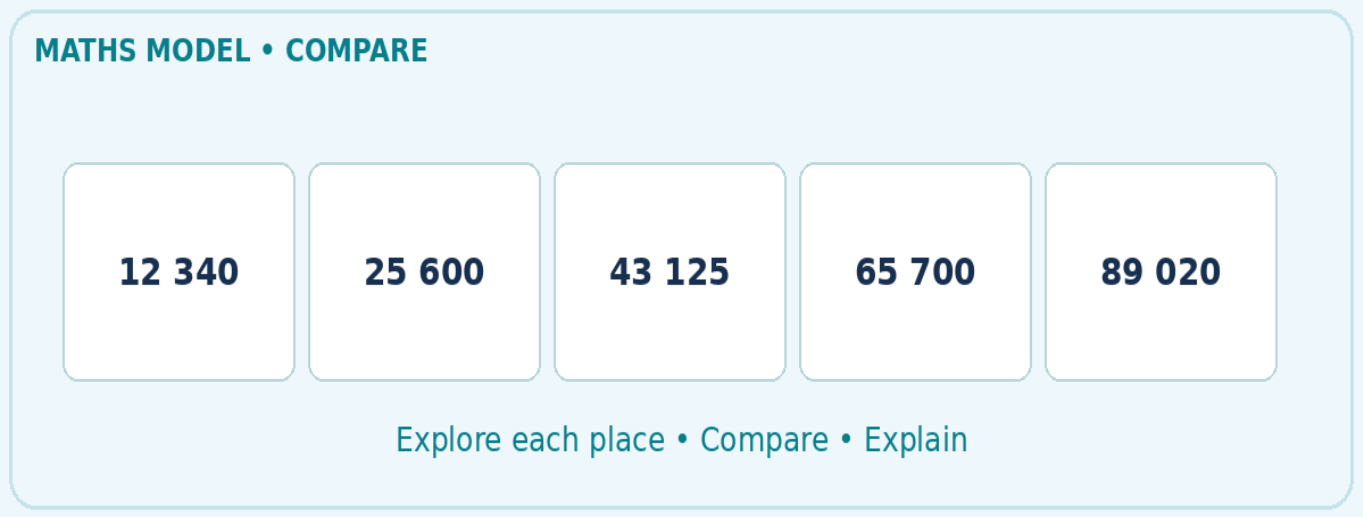
1. 9,956	2. 43,134	3. 26,298
4. 41,558	5. 43,379	6. 49,828
7. 25,786	8. 34,309	9. 18,707
10. 72,751	11. 55,645	12. 61,413
13. 37,377	14. 45,262	15. 25,326

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing and ordering whole numbers

LEARNING GOAL Comparing and ordering whole numbers



LEARN THE IDEA

Compare whole numbers from the highest place value. use < for less than, > for greater than and = for equality.

WORKED EXAMPLE 1

Choose the larger number: 89,830 or 93,149.

Answer: 93,149

Compare the digits from left to right; 93,149 is larger.

WORKED EXAMPLE 2

Which sign makes 90,403 __ 93,995 true?

Answer: <

90,403 is less than 93,995.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Comparing and ordering whole numbers

PLAY · LEARN THE STEPS

1. **Choose the larger number: 19,692 or 17,923.**
   A) 17,923 B) 20,692 C) 19,692 D) 17,823
2. **Which sign makes 22,968 __ 21,031 true?**
   A) none of these B) < C) > D) =
3. **Which sign makes 46,676 __ 47,979 true?**
   A) < B) > C) = D) none of these
4. **Choose the larger number: 76,207 or 76,778.**
   A) 76,778 B) 77,778 C) 76,107 D) 76,207
5. **Which sign makes 45,355 __ 48,890 true?**
   A) < B) > C) = D) none of these

CHALLENGE · DIFFERENT EXAMPLES

1. **Fill the comparison on the scoreboard: 17,566 __ 19,233.**
   A) < B) > C) = D) none of these
2. **Two totals at a science exhibition are 35,142 and 33,339. Which is greater?**
   A) 33,239 B) 33,339 C) 36,142 D) 35,142
3. **Fill the comparison on the scoreboard: 20,674 __ 20,701.**
   A) = B) < C) > D) none of these
4. **Fill the comparison on the scoreboard: 17,028 __ 17,293.**
   A) > B) = C) none of these D) <
5. **Two totals at a farm are 64,309 and 63,708. Which is greater?**
   A) 64,309 B) 63,708 C) 65,309 D) 63,608

Comparing and ordering whole numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two totals during a sports day are 11,229 and 13,994. Which is greater?**
A) 14,994 B) 13,994 C) 11,129 D) 11,229

7. **Fill the comparison on the scoreboard: 29,455 __ 30,234.**
A) < B) > C) none of these D) =

8. **Fill the comparison on the scoreboard: 47,479 __ 49,605.**
A) < B) none of these C) = D) >

9. **Two totals in the class garden are 54,761 and 56,928. Which is greater?**
A) 54,761 B) 57,928 C) 56,928 D) 54,661

10. **Fill the comparison on the scoreboard: 39,742 __ 37,715.**
A) < B) none of these C) = D) >

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 19,692	2. >	3. <
4. 76,778	5. <	6. <
7. 35,142	8. <	9. <
10. 64,309	11. 13,994	12. <
13. <	14. 56,928	15. >

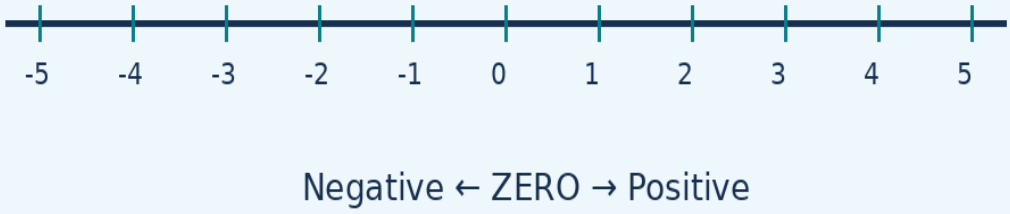
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Rounding large whole numbers

LEARNING GOAL Rounding large whole numbers

MATHS MODEL • INTEGER



Negative ← ZERO → Positive

LEARN THE IDEA

To round, find the target place and inspect the digit immediately to its right. five or more rounds up.

WORKED EXAMPLE 1

Round 55,994 to the nearest 100.

Answer: 56,000

Check the next digit and round 55,994 to 56,000.

WORKED EXAMPLE 2

Round 40,606 to the nearest 10.

Answer: 40,610

Check the next digit and round 40,606 to 40,610.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Rounding large whole numbers

PLAY · LEARN THE STEPS

1. **Round 87,600 to the nearest 10.**
   A) 87,590 B) 87,600 C) 87601 D) 87,610
2. **Round 17,674 to the nearest 10,000.**
   A) 10,000 B) 30,000 C) 17,674 D) 20,000
3. **Round 90,034 to the nearest 10,000.**
   A) 90,000 B) 90,034 C) 100,000 D) 80,000
4. **Round 16,538 to the nearest 10.**
   A) 16,530 B) 16,550 C) 16,538 D) 16,540
5. **Round 33,987 to the nearest 100.**
   A) 34,100 B) 33,987 C) 34,000 D) 33,900

CHALLENGE · DIFFERENT EXAMPLES

1. **A crowd estimate at a science exhibition is based on 54,130 visitors. To the nearest 1,000, what should be reported?**
   A) 54,000 B) 55,000 C) 54,130 D) 53,000
2. **A crowd estimate during a reading competition is based on 85,011 visitors. To the nearest 1,000, what should be reported?**
   A) 85,000 B) 84,000 C) 86,000 D) 85,011
3. **A crowd estimate for a school fundraiser is based on 16,658 visitors. To the nearest 100, what should be reported?**
   A) 16,800 B) 16,700 C) 16,658 D) 16,600
4. **A crowd estimate at a farm is based on 76,562 visitors. To the nearest 100, what should be reported?**
   A) 76,562 B) 76,600 C) 76,700 D) 76,500
5. **A crowd estimate during a sports day is based on 17,176 visitors. To the nearest 100, what should be reported?**
   A) 17,200 B) 17,176 C) 17,100 D) 17,300

Rounding large whole numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **A crowd estimate in a craft workshop is based on 87,739 visitors. To the nearest 10, what should be reported?**

A) 87,750 B) 87,739 C) 87,740 D) 87,730

7 **A crowd estimate at a school fair is based on 10,906 visitors. To the nearest 10,000, what should be reported?**

A) 0 B) 20,000 C) 10,906 D) 10,000

8 **A crowd estimate in the class garden is based on 42,948 visitors. To the nearest 10,000, what should be reported?**

A) 30,000 B) 50,000 C) 40,000 D) 42,948

9 **A crowd estimate at a market stall is based on 53,979 visitors. To the nearest 1,000, what should be reported?**

A) 54,000 B) 55,000 C) 53,979 D) 53,000

10 **A crowd estimate during a library project is based on 70,269 visitors. To the nearest 10, what should be reported?**

A) 70,270 B) 70,280 C) 70,260 D) 70,269

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 87,600	2. 20,000	3. 90,000
4. 16,540	5. 34,000	6. 54,000
7. 85,000	8. 16,700	9. 76,600
10. 17,200	11. 87,740	12. 10,000
13. 40,000	14. 54,000	15. 70,270

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Skip counting forwards and backwards

LEARNING GOAL Skip counting forwards and backwards



LEARN THE IDEA

A skip-counting pattern adds or subtracts the same step at every landing. check both forward and backward travel.

WORKED EXAMPLE 1

Count 4 jumps forward in 2,000s from 42,000.

Answer: 50,000

42,000 + 4 × 2,000 = 50,000.

WORKED EXAMPLE 2

Count 3 jumps forward in 2,000s from 34,000.

Answer: 40,000

34,000 + 3 × 2,000 = 40,000.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Skip counting forwards and backwards

PLAY · LEARN THE STEPS

1. **Count 5 jumps back in 2,000s from 50,000.**
   A) 38,000 B) 60,000 C) 42,000 D) 40,000
2. **Count 6 jumps forward in 1,000s from 34,000.**
   A) 40,000 B) 41,000 C) 28,000 D) 39,000
3. **Count 6 jumps forward in 2,000s from 36,000.**
   A) 24,000 B) 46,000 C) 48,000 D) 50,000
4. **Count 4 jumps forward in 1,000s from 12,000.**
   A) 15,000 B) 17,000 C) 16,000 D) 8,000
5. **Count 5 jumps back in 2,500s from 47,500.**
   A) 60,000 B) 37,500 C) 35,000 D) 32,500

CHALLENGE · DIFFERENT EXAMPLES

1. **A route during a reading competition moves 4 jumps of 1,000 backward from 19,000. Where does it finish?**
   A) 15,000 B) 16,000 C) 23,000 D) 14,000
2. **A route for a school fundraiser moves 5 jumps of 2,000 backward from 66,000. Where does it finish?**
   A) 76,000 B) 54,000 C) 56,000 D) 58,000
3. **A route at a farm moves 6 jumps of 1,000 forward from 23,000. Where does it finish?**
   A) 30,000 B) 29,000 C) 17,000 D) 28,000
4. **A route during a sports day moves 2 jumps of 1,000 forward from 18,000. Where does it finish?**
   A) 20,000 B) 16,000 C) 21,000 D) 19,000
5. **A route in a craft workshop moves 5 jumps of 2,500 forward from 30,000. Where does it finish?**
   A) 45,000 B) 17,500 C) 42,500 D) 40,000

Skip counting forwards and backwards

CHALLENGE · APPLY WHAT YOU KNOW

6 **A route at a school fair moves 6 jumps of 2,000 backward from 32,000. Where does it finish?**

A) 44,000 B) 18,000 C) 20,000 D) 22,000

7 **A route in the class garden moves 3 jumps of 500 backward from 13,000. Where does it finish?**

A) 14,500 B) 11,000 C) 12,000 D) 11,500

8 **A route at a market stall moves 3 jumps of 1,000 forward from 17,000. Where does it finish?**

A) 14,000 B) 20,000 C) 21,000 D) 19,000

9 **A route during a library project moves 6 jumps of 500 backward from 9,500. Where does it finish?**

A) 6,000 B) 12,500 C) 7,000 D) 6,500

10 **A route on a football field moves 2 jumps of 2,500 forward from 37,500. Where does it finish?**

A) 40,000 B) 32,500 C) 42,500 D) 45,000

ANSWER CHECK · LOOK ONLY AFTER TRYING

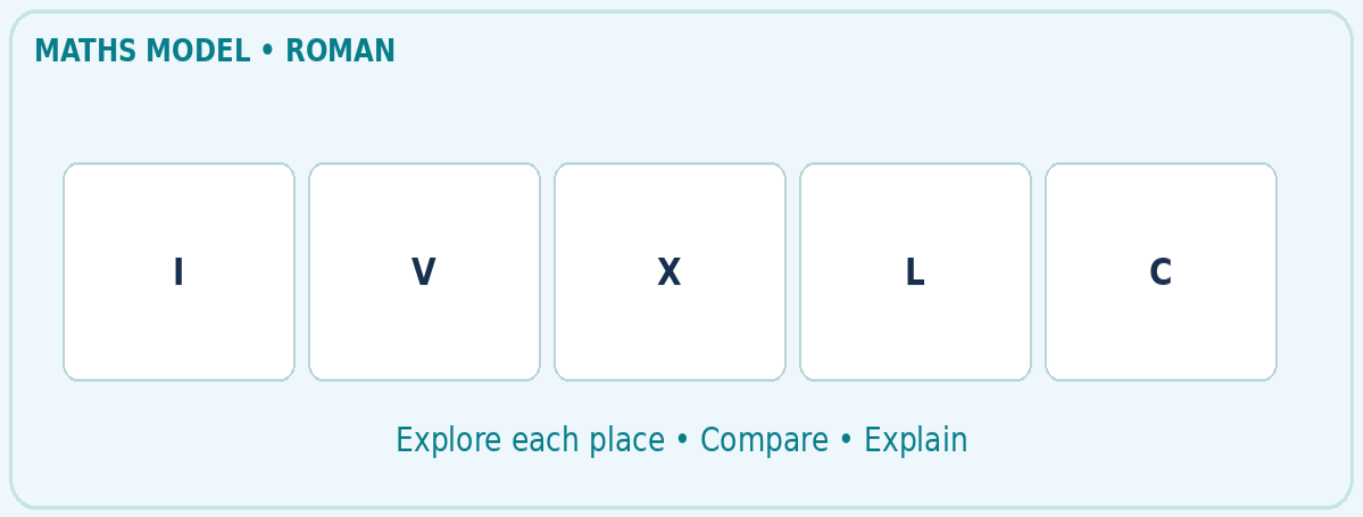
1. 40,000	2. 40,000	3. 48,000
4. 16,000	5. 35,000	6. 15,000
7. 56,000	8. 29,000	9. 20,000
10. 42,500	11. 20,000	12. 11,500
13. 20,000	14. 6,500	15. 42,500

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Roman numeral symbols up to C

LEARNING GOAL Roman numeral symbols up to C



LEARN THE IDEA

Roman symbols include i=1, v=5, x=10, l=50 and c=100. use iv, ix, xl and xc for subtraction forms.

WORKED EXAMPLE 1

Write 11 as a Roman numeral.

Answer: XI

Roman numerals use I, V, X, L and C: 11 = XI.

WORKED EXAMPLE 2

Convert LXXXIII into an ordinary number.

Answer: 83

LXXXIII represents 83.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Roman numeral symbols up to C

PLAY · LEARN THE STEPS

1. **Write 29 as a Roman numeral.**
   A) XXXVII B) XLVIII C) LXVI D) XXIX
2. **Convert LXIII into an ordinary number.**
   A) 63 B) 58 C) 73 D) 68
3. **Write 27 as a Roman numeral.**
   A) XLVI B) LXIV C) XXXV D) XXVII
4. **Convert LII into an ordinary number.**
   A) 52 B) 62 C) 47 D) 57
5. **Write 46 as a Roman numeral.**
   A) XLVI B) LXV C) LXXXIII D) LIV

CHALLENGE · DIFFERENT EXAMPLES

1. **A clock label for a school fundraiser is XLVII. What number is it?**
   A) 42 B) 52 C) 47 D) 57
2. **The award board at a farm shows position 24. Which Roman numeral is correct?**
   A) XLIII B) XXIV C) XXXII D) LXI
3. **A clock label during a sports day is XXXV. What number is it?**
   A) 45 B) 35 C) 30 D) 40
4. **The award board in a craft workshop shows position 34. Which Roman numeral is correct?**
   A) XLII B) LXXI C) LIII D) XXXIV
5. **A clock label at a school fair is XXV. What number is it?**
   A) 20 B) 25 C) 35 D) 30

Roman numeral symbols up to C

CHALLENGE · APPLY WHAT YOU KNOW

6. **The award board in the class garden shows position 91. Which Roman numeral is correct?**
   A) X B) XXVIII C) XCIX D) XCI

7. **A clock label at a market stall is VIII. What number is it?**
   A) 18 B) 13 C) 8 D) 3

8. **The award board during a library project shows position 79. Which Roman numeral is correct?**
   A) LXXXVII B) XCVIII C) XVI D) LXXIX

9. **A clock label on a football field is LXXXIII. What number is it?**
   A) 93 B) 88 C) 78 D) 83

10. **The award board in a recycling club shows position 46. Which Roman numeral is correct?**
    A) LIV B) XLVI C) LXXXIII D) LXV

ANSWER CHECK · LOOK ONLY AFTER TRYING

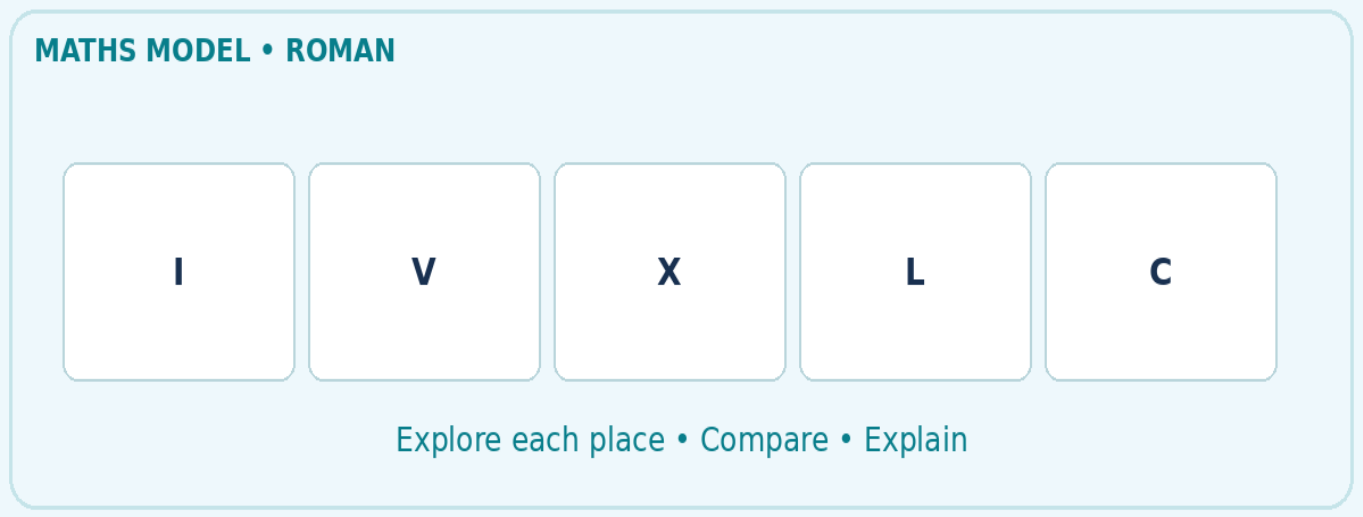
1. XXIX	2. 63	3. XXVII
4. 52	5. XLVI	6. 47
7. XXIV	8. 35	9. XXXIV
10. 25	11. XCI	12. 8
13. LXXIX	14. 83	15. XLVI

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Converting Roman and Arabic numerals

LEARNING GOAL Converting Roman and Arabic numerals



LEARN THE IDEA

Roman symbols include i=1, v=5, x=10, l=50 and c=100. use iv, ix, xl and xc for subtraction forms.

WORKED EXAMPLE 1

Write 64 as a Roman numeral.

Answer: LXIV

Roman numerals use I, V, X, L and C: 64 = LXIV.

WORKED EXAMPLE 2

Convert XXXIV into an ordinary number.

Answer: 34

XXXIV represents 34.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Converting Roman and Arabic numerals

PLAY · LEARN THE STEPS

1. **Write 40 as a Roman numeral.**
   A) LIX B) XL C) LXXVII D) XLVIII
2. **Convert XXVII into an ordinary number.**
   A) 22 B) 27 C) 37 D) 32
3. **Write 12 as a Roman numeral.**
   A) XXXI B) XLIX C) XX D) XII
4. **Convert XCV into an ordinary number.**
   A) 90 B) 105 C) 95 D) 100
5. **Write 23 as a Roman numeral.**
   A) XXXI B) XXIII C) LX D) XLII

CHALLENGE · DIFFERENT EXAMPLES

1. **A clock label at a farm is XLII. What number is it?**
   A) 52 B) 37 C) 42 D) 47
2. **The award board during a sports day shows position 2. Which Roman numeral is correct?**
   A) II B) XXXIX C) XXI D) X
3. **A clock label in a craft workshop is XXXVI. What number is it?**
   A) 46 B) 41 C) 31 D) 36
4. **The award board at a school fair shows position 43. Which Roman numeral is correct?**
   A) XLIII B) LXII C) LI D) LXXX
5. **A clock label in the class garden is XLV. What number is it?**
   A) 55 B) 50 C) 40 D) 45

Converting Roman and Arabic numerals

CHALLENGE · APPLY WHAT YOU KNOW

6. **The award board at a market stall shows position 21. Which Roman numeral is correct?**
   A) XL B) LVIII C) XXIX D) XXI

7. **A clock label during a library project is LII. What number is it?**
   A) 62 B) 47 C) 52 D) 57

8. **The award board on a football field shows position 76. Which Roman numeral is correct?**
   A) XCV B) XIII C) LXXVI D) LXXXIV

9. **A clock label in a recycling club is L. What number is it?**
   A) 55 B) 60 C) 50 D) 45

10. **The award board at a science exhibition shows position 31. Which Roman numeral is correct?**
    A) XXXIX B) LXVIII C) XXXI D) L

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. XL	2. 27	3. XII
4. 95	5. XXIII	6. 42
7. II	8. 36	9. XLIII
10. 45	11. XXI	12. 52
13. LXXVI	14. 50	15. XXXI

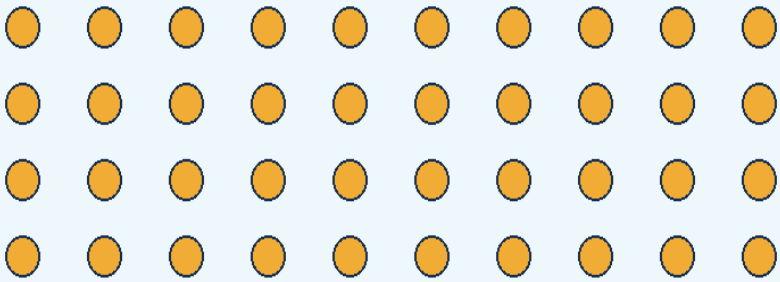
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding factors up to 100

LEARNING GOAL Finding factors up to 100

MATHS MODEL • FACTORS



4 groups of 10 = 40

LEARN THE IDEA

A factor divides a number into equal groups without a remainder. multiplication and division are inverse operations.

WORKED EXAMPLE 1

How many groups of 2 are in 12?

Answer: 6

12 ÷ 2 = 6; 2 is a factor of 12.

WORKED EXAMPLE 2

How many groups of 8 are in 24?

Answer: 3

24 ÷ 8 = 3; 8 is a factor of 24.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Finding factors up to 100

PLAY · LEARN THE STEPS

1. **How many groups of 2 are in 18?**
   A) 11 B) 2 C) 18 D) 9
2. **How many groups of 24 are in 24?**
   A) 1 B) 3 C) 2 D) 24
3. **How many groups of 8 are in 24?**
   A) 24 B) 3 C) 8 D) 5
4. **How many groups of 8 are in 40?**
   A) 8 B) 40 C) 7 D) 5
5. **How many groups of 7 are in 35?**
   A) 7 B) 5 C) 35 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A teacher during a sports day arranges 27 objects in 9 equal rows. How many objects are in each row?**
   A) 9 B) 3 C) 5 D) 27
2. **A teacher in a craft workshop arranges 24 objects in 12 equal rows. How many objects are in each row?**
   A) 2 B) 12 C) 24 D) 4
3. **A teacher at a school fair arranges 16 objects in 2 equal rows. How many objects are in each row?**
   A) 8 B) 2 C) 10 D) 16
4. **A teacher in the class garden arranges 10 objects in 10 equal rows. How many objects are in each row?**
   A) 3 B) 10 C) 1 D) 2
5. **A teacher at a market stall arranges 40 objects in 20 equal rows. How many objects are in each row?**
   A) 40 B) 2 C) 20 D) 4

Finding factors up to 100

CHALLENGE · APPLY WHAT YOU KNOW

6 **A teacher during a library project arranges 56 objects in 7 equal rows. How many objects are in each row?**

A) 10 B) 7 C) 56 D) 8

7 **A teacher on a football field arranges 36 objects in 6 equal rows. How many objects are in each row?**

A) 6 B) 36 C) 7 D) 8

8 **A teacher in a recycling club arranges 49 objects in 7 equal rows. How many objects are in each row?**

A) 7 B) 8 C) 9 D) 49

9 **A teacher at a science exhibition arranges 54 objects in 54 equal rows. How many objects are in each row?**

A) 3 B) 1 C) 2 D) 54

10 **A teacher during a reading competition arranges 49 objects in 7 equal rows. How many objects are in each row?**

A) 8 B) 49 C) 7 D) 9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 9	2. 1	3. 3
4. 5	5. 5	6. 3
7. 2	8. 8	9. 1
10. 2	11. 8	12. 6
13. 7	14. 1	15. 7

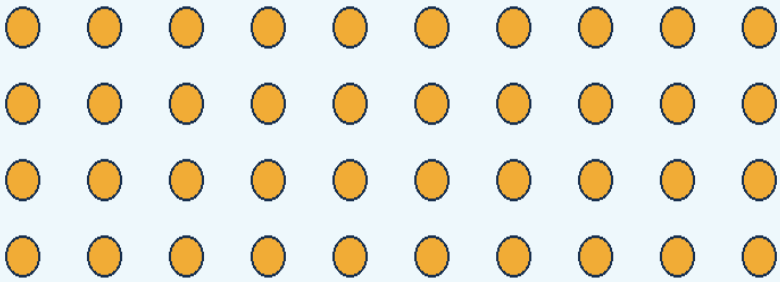
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Prime and composite numbers

LEARNING GOAL Prime and composite numbers

MATHS MODEL • PRIME



4 groups of 10 = 40

LEARN THE IDEA

A prime number has exactly two factors: one and itself. a composite number has more than two factors.

WORKED EXAMPLE 1

Is 9 prime or composite?

Answer: composite

9 has more than two factors.

WORKED EXAMPLE 2

Is 37 prime or composite?

Answer: prime

37 has only 1 and itself.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Prime and composite numbers

PLAY · LEARN THE STEPS

1. **Is 49 prime or composite?**
   A) even only B) composite C) prime D) neither
2. **Is 2 prime or composite?**
   A) composite B) prime C) even only D) neither
3. **Is 96 prime or composite?**
   A) neither B) composite C) even only D) prime
4. **Is 83 prime or composite?**
   A) prime B) neither C) composite D) even only
5. **Is 74 prime or composite?**
   A) neither B) even only C) prime D) composite

CHALLENGE · DIFFERENT EXAMPLES

1. **A number card in a craft workshop contains 43. Which category describes it?**
   A) even only B) prime C) composite D) neither
2. **A number card at a school fair contains 25. Which category describes it?**
   A) neither B) even only C) composite D) prime
3. **A number card in the class garden contains 61. Which category describes it?**
   A) prime B) even only C) neither D) composite
4. **A number card at a market stall contains 21. Which category describes it?**
   A) prime B) even only C) neither D) composite
5. **A number card during a library project contains 47. Which category describes it?**
   A) composite B) neither C) even only D) prime

Prime and composite numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **A number card on a football field contains 35. Which category describes it?**
A) even only B) neither C) prime D) composite

7. **A number card in a recycling club contains 73. Which category describes it?**
A) neither B) composite C) even only D) prime

8. **A number card at a science exhibition contains 50. Which category describes it?**
A) composite B) neither C) even only D) prime

9. **A number card during a reading competition contains 17. Which category describes it?**
A) even only B) prime C) neither D) composite

10. **A number card for a school fundraiser contains 51. Which category describes it?**
A) prime B) composite C) even only D) neither

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. composite	2. prime	3. composite
4. prime	5. composite	6. prime
7. composite	8. prime	9. composite
10. prime	11. composite	12. prime
13. composite	14. prime	15. composite

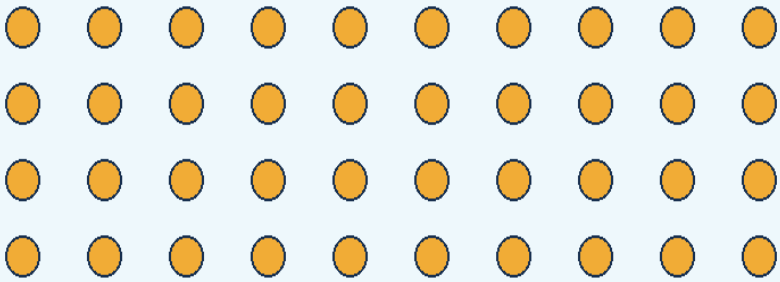
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Even and odd numbers

LEARNING GOAL Even and odd numbers

MATHS MODEL • ODD



4 groups of 10 = 40

LEARN THE IDEA

An even number can be put into pairs without leftovers; an odd number leaves one when arranged in pairs.

WORKED EXAMPLE 1

Is 94 even or odd?

Answer: even

94 can be split into equal groups of two.

WORKED EXAMPLE 2

Is 61 even or odd?

Answer: odd

61 cannot be split into equal groups of two.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Even and odd numbers

PLAY · LEARN THE STEPS

1. **Is 80 even or odd?**
   A) odd B) even C) multiple of five D) prime
2. **Is 61 even or odd?**
   A) even B) multiple of five C) prime D) odd
3. **Is 98 even or odd?**
   A) prime B) even C) multiple of five D) odd
4. **Is 71 even or odd?**
   A) even B) multiple of five C) odd D) prime
5. **Is 40 even or odd?**
   A) odd B) multiple of five C) prime D) even

CHALLENGE · DIFFERENT EXAMPLES

1. **A bag at a school fair contains 49 marbles. Could they be paired without a leftover? Choose the number type.**
   A) multiple of five B) odd C) prime D) even
2. **A bag in the class garden contains 92 marbles. Could they be paired without a leftover? Choose the number type.**
   A) odd B) even C) prime D) multiple of five
3. **A bag at a market stall contains 31 marbles. Could they be paired without a leftover? Choose the number type.**
   A) even B) odd C) multiple of five D) prime
4. **A bag during a library project contains 84 marbles. Could they be paired without a leftover? Choose the number type.**
   A) odd B) even C) prime D) multiple of five
5. **A bag on a football field contains 33 marbles. Could they be paired without a leftover? Choose the number type.**
   A) prime B) even C) odd D) multiple of five

Even and odd numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **A bag in a recycling club contains 80 marbles. Could they be paired without a leftover? Choose the number type.**

A) even B) multiple of five C) odd D) prime

7 **A bag at a science exhibition contains 63 marbles. Could they be paired without a leftover? Choose the number type.**

A) prime B) odd C) even D) multiple of five

8 **A bag during a reading competition contains 70 marbles. Could they be paired without a leftover? Choose the number type.**

A) multiple of five B) even C) prime D) odd

9 **A bag for a school fundraiser contains 33 marbles. Could they be paired without a leftover? Choose the number type.**

A) prime B) multiple of five C) even D) odd

10 **A bag at a farm contains 70 marbles. Could they be paired without a leftover? Choose the number type.**

A) multiple of five B) even C) prime D) odd

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. even	2. odd	3. even
4. odd	5. even	6. odd
7. even	8. odd	9. even
10. odd	11. even	12. odd
13. even	14. odd	15. even

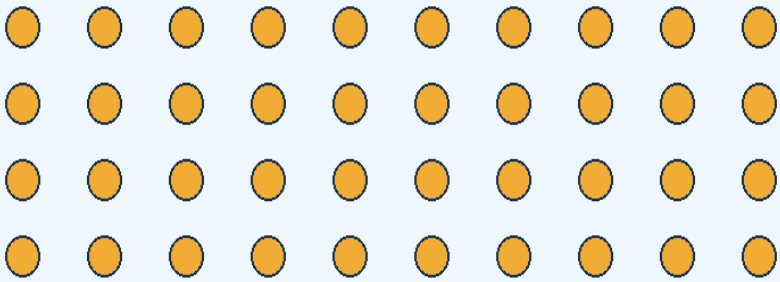
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Highest common factors

LEARNING GOAL Highest common factors

MATHS MODEL • HCF



4 groups of 10 = 40

LEARN THE IDEA

The highest common factor is the largest number that divides both numbers. factor trees and prime factors help.

WORKED EXAMPLE 1

Find HCF(6, 15).

Answer: 3

Highest common factor = 3.

WORKED EXAMPLE 2

Find HCF(24, 10).

Answer: 2

Highest common factor = 2.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Highest common factors

PLAY · LEARN THE STEPS

1. **Find HCF(12, 40).**
   A) 52 B) 4 C) 3 D) 5
2. **Find HCF(9, 28).**
   A) 1 B) 2 C) 3 D) 37
3. **Find HCF(21, 30).**
   A) 2 B) 4 C) 3 D) 51
4. **Find HCF(32, 30).**
   A) 2 B) 3 C) 62 D) 1
5. **Find HCF(40, 35).**
   A) 75 B) 5 C) 4 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **Two groups in the class garden hold 20 and 6 items. What is the greatest size of identical packets that divides both totals?**
   A) 26 B) 1 C) 2 D) 3
2. **Two groups at a market stall hold 60 and 24 items. What is the greatest size of identical packets that divides both totals?**
   A) 13 B) 12 C) 11 D) 84
3. **Two groups during a library project hold 24 and 4 items. What is the greatest size of identical packets that divides both totals?**
   A) 28 B) 3 C) 4 D) 5
4. **Two groups on a football field hold 20 and 24 items. What is the greatest size of identical packets that divides both totals?**
   A) 44 B) 4 C) 5 D) 3
5. **Two groups in a recycling club hold 6 and 12 items. What is the greatest size of identical packets that divides both totals?**
   A) 5 B) 6 C) 7 D) 18

Highest common factors

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two groups at a science exhibition hold 24 and 27 items. What is the greatest size of identical packets that divides both totals?**

A) 51 B) 4 C) 3 D) 2

7 **Two groups during a reading competition hold 27 and 12 items. What is the greatest size of identical packets that divides both totals?**

A) 3 B) 2 C) 39 D) 4

8 **Two groups for a school fundraiser hold 24 and 20 items. What is the greatest size of identical packets that divides both totals?**

A) 4 B) 5 C) 3 D) 44

9 **Two groups at a farm hold 18 and 10 items. What is the greatest size of identical packets that divides both totals?**

A) 3 B) 28 C) 2 D) 1

10 **Two groups during a sports day hold 6 and 18 items. What is the greatest size of identical packets that divides both totals?**

A) 5 B) 7 C) 6 D) 24

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4	2. 1	3. 3
4. 2	5. 5	6. 2
7. 12	8. 4	9. 4
10. 6	11. 3	12. 3
13. 4	14. 2	15. 6

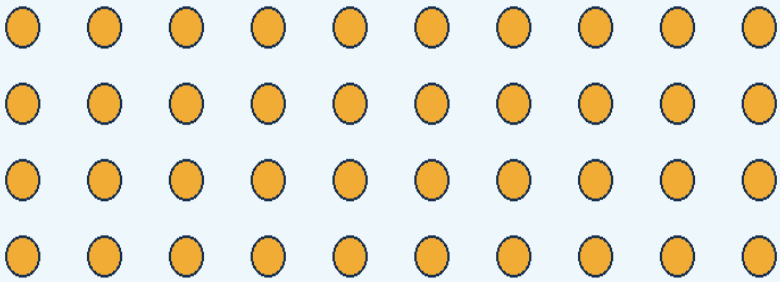
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiples and factor relationships

LEARNING GOAL Multiples and factor relationships

MATHS MODEL • LCM



4 groups of 10 = 40

LEARN THE IDEA

Multiples appear when you count by a number. the least common multiple is the first shared positive multiple.

WORKED EXAMPLE 1

Find LCM(9, 10).

Answer: 90

Least common multiple = 90.

WORKED EXAMPLE 2

Find LCM(11, 2).

Answer: 22

Least common multiple = 22.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multiples and factor relationships

PLAY · LEARN THE STEPS

1. **Find LCM(8, 6).**
   A) 24 B) 25 C) 23 D) 14
2. **Find LCM(12, 10).**
   A) 59 B) 60 C) 22 D) 61
3. **Find LCM(8, 7).**
   A) 55 B) 15 C) 57 D) 56
4. **Find LCM(11, 5).**
   A) 16 B) 54 C) 56 D) 55
5. **Find LCM(9, 2).**
   A) 17 B) 19 C) 18 D) 11

CHALLENGE · DIFFERENT EXAMPLES

1. **Two bells at a market stall ring every 4 and 10 minutes. After how many minutes will they ring together?**
   A) 19 B) 20 C) 14 D) 21
2. **Two bells during a library project ring every 12 and 4 minutes. After how many minutes will they ring together?**
   A) 16 B) 12 C) 11 D) 13
3. **Two bells on a football field ring every 10 and 3 minutes. After how many minutes will they ring together?**
   A) 13 B) 31 C) 29 D) 30
4. **Two bells in a recycling club ring every 6 and 8 minutes. After how many minutes will they ring together?**
   A) 14 B) 24 C) 23 D) 25
5. **Two bells at a science exhibition ring every 8 and 10 minutes. After how many minutes will they ring together?**
   A) 41 B) 39 C) 18 D) 40

Multiples and factor relationships

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two bells during a reading competition ring every 4 and 5 minutes. After how many minutes will they ring together?**

A) 19 B) 20 C) 21 D) 9

7 **Two bells for a school fundraiser ring every 9 and 2 minutes. After how many minutes will they ring together?**

A) 17 B) 11 C) 18 D) 19

8 **Two bells at a farm ring every 9 and 6 minutes. After how many minutes will they ring together?**

A) 17 B) 18 C) 15 D) 19

9 **Two bells during a sports day ring every 6 and 7 minutes. After how many minutes will they ring together?**

A) 13 B) 43 C) 42 D) 41

10 **Two bells in a craft workshop ring every 5 and 10 minutes. After how many minutes will they ring together?**

A) 11 B) 10 C) 15 D) 9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 24	2. 60	3. 56
4. 55	5. 18	6. 20
7. 12	8. 30	9. 24
10. 40	11. 20	12. 18
13. 18	14. 42	15. 10

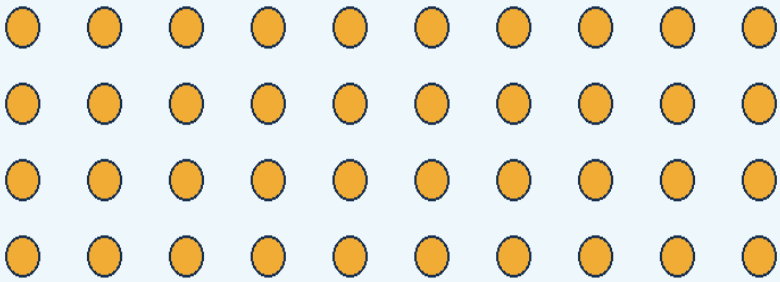
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Mental strategies for multiplication facts

LEARNING GOAL Mental strategies for multiplication facts

MATHS MODEL • MENTAL FACTS



4 groups of 10 = 40

LEARN THE IDEA

Build multiplication facts from known facts. for example, 6×7 = 5×7 + 7, so 35 + 7 = 42.

WORKED EXAMPLE 1

Use the known fact 1 × 3 = 3 to find 2 × 3.

Answer: 6

Add one group of 3: 3+3=6.

WORKED EXAMPLE 2

Use the known fact 5 × 5 = 25 to find 6 × 5.

Answer: 30

Add one group of 5: 25+5=30.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Mental strategies for multiplication facts

PLAY · LEARN THE STEPS

1. **Use the known fact 8 × 5 = 40 to find 9 × 5.**
   A) 14 B) 45 C) 40 D) 50
2. **Use the known fact 5 × 6 = 30 to find 6 × 6.**
   A) 30 B) 12 C) 42 D) 36
3. **Use the known fact 4 × 3 = 12 to find 5 × 3.**
   A) 15 B) 18 C) 8 D) 12
4. **Use the known fact 8 × 2 = 16 to find 9 × 2.**
   A) 18 B) 20 C) 16 D) 11
5. **Use the known fact 3 × 5 = 15 to find 4 × 5.**
   A) 15 B) 9 C) 25 D) 20

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner during a library project knows 4 × 5 = 20. How can they calculate 5 × 5 using one more group?**
   A) 30 B) 20 C) 25 D) 10
2. **A learner on a football field knows 5 × 3 = 15. How can they calculate 6 × 3 using one more group?**
   A) 9 B) 21 C) 18 D) 15
3. **A learner in a recycling club knows 3 × 7 = 21. How can they calculate 4 × 7 using one more group?**
   A) 28 B) 11 C) 35 D) 21
4. **A learner at a science exhibition knows 3 × 8 = 24. How can they calculate 4 × 8 using one more group?**
   A) 40 B) 24 C) 12 D) 32
5. **A learner during a reading competition knows 6 × 3 = 18. How can they calculate 7 × 3 using one more group?**
   A) 10 B) 21 C) 18 D) 24

Mental strategies for multiplication facts

CHALLENGE · APPLY WHAT YOU KNOW

6. **A learner for a school fundraiser knows 8 × 5 = 40. How can they calculate 9 × 5 using one more group?**
   A) 50 B) 40 C) 45 D) 14

7. **A learner at a farm knows 8 × 2 = 16. How can they calculate 9 × 2 using one more group?**
   A) 16 B) 11 C) 18 D) 20

8. **A learner during a sports day knows 1 × 3 = 3. How can they calculate 2 × 3 using one more group?**
   A) 3 B) 6 C) 9 D) 5

9. **A learner in a craft workshop knows 6 × 2 = 12. How can they calculate 7 × 2 using one more group?**
   A) 9 B) 14 C) 16 D) 12

10. **A learner at a school fair knows 8 × 8 = 64. How can they calculate 9 × 8 using one more group?**
    A) 64 B) 72 C) 80 D) 17

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 45	2. 36	3. 15
4. 18	5. 20	6. 25
7. 18	8. 28	9. 32
10. 21	11. 45	12. 18
13. 6	14. 14	15. 72

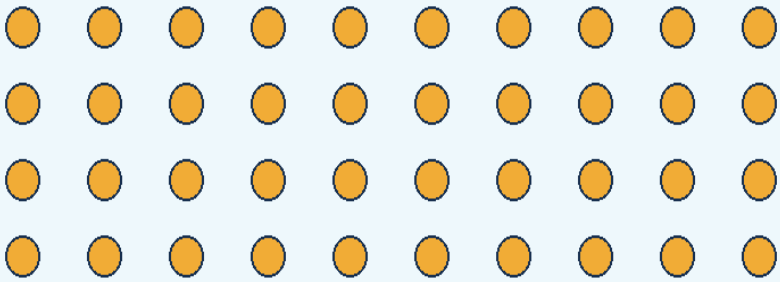
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Efficient mental multiplication

LEARNING GOAL Efficient mental multiplication

MATHS MODEL • MENTAL MULT



4 groups of 10 = 40

LEARN THE IDEA

Use place value and doubling: multiplying by 20 is multiplying by 2 and then by 10.

WORKED EXAMPLE 1

Work out 64 × 100 using an efficient method.

Answer: 6400

64 × 100 = 6400.

WORKED EXAMPLE 2

Work out 40 × 9 using an efficient method.

Answer: 360

40 × 9 = 360.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Efficient mental multiplication

PLAY · LEARN THE STEPS

1. **Work out 94 × 20 using an efficient method.**
   A) 1880 B) 1890 C) 1900 D) 1860
2. **Work out 23 × 20 using an efficient method.**
   A) 440 B) 470 C) 480 D) 460
3. **Work out 58 × 9 using an efficient method.**
   A) 531 B) 513 C) 522 D) 532
4. **Work out 24 × 20 using an efficient method.**
   A) 500 B) 460 C) 480 D) 490
5. **Work out 99 × 20 using an efficient method.**
   A) 1990 B) 1960 C) 1980 D) 2000

CHALLENGE · DIFFERENT EXAMPLES

1. **A project on a football field has 44 units in each of 5 sets. What is the total?**
   A) 220 B) 225 C) 230 D) 215
2. **A project in a recycling club has 67 units in each of 100 sets. What is the total?**
   A) 6710 B) 6800 C) 6700 D) 6600
3. **A project at a science exhibition has 38 units in each of 5 sets. What is the total?**
   A) 190 B) 195 C) 200 D) 185
4. **A project during a reading competition has 60 units in each of 100 sets. What is the total?**
   A) 6000 B) 6100 C) 5900 D) 6010
5. **A project for a school fundraiser has 70 units in each of 20 sets. What is the total?**
   A) 1380 B) 1400 C) 1410 D) 1420

Efficient mental multiplication

CHALLENGE · APPLY WHAT YOU KNOW

6. **A project at a farm has 57 units in each of 100 sets. What is the total?**
A) 5800 B) 5600 C) 5700 D) 5710

7. **A project during a sports day has 83 units in each of 10 sets. What is the total?**
A) 820 B) 840 C) 831 D) 830

8. **A project in a craft workshop has 33 units in each of 20 sets. What is the total?**
A) 670 B) 680 C) 640 D) 660

9. **A project at a school fair has 95 units in each of 100 sets. What is the total?**
A) 9600 B) 9510 C) 9400 D) 9500

10. **A project in the class garden has 34 units in each of 9 sets. What is the total?**
A) 306 B) 316 C) 297 D) 315

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 1880	2. 460	3. 522
4. 480	5. 1980	6. 220
7. 6700	8. 190	9. 6000
10. 1400	11. 5700	12. 830
13. 660	14. 9500	15. 306

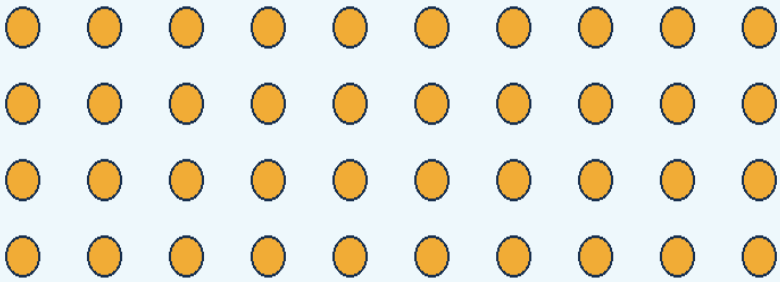
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplying by two-digit numbers

LEARNING GOAL Multiplying by two-digit numbers

MATHS MODEL • MULTIPLY



4 groups of 10 = 40

LEARN THE IDEA

Multiply each place of one number by the other. add partial products and check the place-value columns.

WORKED EXAMPLE 1

Work out 256 × 43 using an efficient method.

Answer: 11008

256 × 43 = 11008.

WORKED EXAMPLE 2

Work out 290 × 16 using an efficient method.

Answer: 4640

290 × 16 = 4640.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multiplying by two-digit numbers

PLAY · LEARN THE STEPS

1. **Work out 402 × 20 using an efficient method.**
   A) 8060 B) 8050 C) 8020 D) 8040
2. **Work out 475 × 41 using an efficient method.**
   A) 19516 B) 19485 C) 19434 D) 19475
3. **Work out 393 × 41 using an efficient method.**
   A) 16113 B) 16123 C) 16072 D) 16154
4. **Work out 537 × 34 using an efficient method.**
   A) 18268 B) 18292 C) 18258 D) 18224
5. **Work out 370 × 27 using an efficient method.**
   A) 9990 B) 10000 C) 10017 D) 9963

CHALLENGE · DIFFERENT EXAMPLES

1. **A project in a recycling club has 801 units in each of 44 sets. What is the total?**
   A) 35254 B) 35244 C) 35288 D) 35200
2. **A project at a science exhibition has 282 units in each of 42 sets. What is the total?**
   A) 11844 B) 11854 C) 11802 D) 11886
3. **A project during a reading competition has 800 units in each of 15 sets. What is the total?**
   A) 12015 B) 12000 C) 12010 D) 11985
4. **A project for a school fundraiser has 674 units in each of 14 sets. What is the total?**
   A) 9450 B) 9446 C) 9436 D) 9422
5. **A project at a farm has 280 units in each of 42 sets. What is the total?**
   A) 11770 B) 11718 C) 11760 D) 11802

Multiplying by two-digit numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **A project during a sports day has 758 units in each of 37 sets. What is the total?**
A) 28056 B) 28009 C) 28083 D) 28046

7. **A project in a craft workshop has 154 units in each of 20 sets. What is the total?**
A) 3080 B) 3090 C) 3060 D) 3100

8. **A project at a school fair has 179 units in each of 28 sets. What is the total?**
A) 5012 B) 4984 C) 5040 D) 5022

9. **A project in the class garden has 680 units in each of 47 sets. What is the total?**
A) 31960 B) 32007 C) 31970 D) 31913

10. **A project at a market stall has 279 units in each of 22 sets. What is the total?**
A) 6160 B) 6116 C) 6138 D) 6148

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 8040	2. 19475	3. 16113
4. 18258	5. 9990	6. 35244
7. 11844	8. 12000	9. 9436
10. 11760	11. 28046	12. 3080
13. 5012	14. 31960	15. 6138

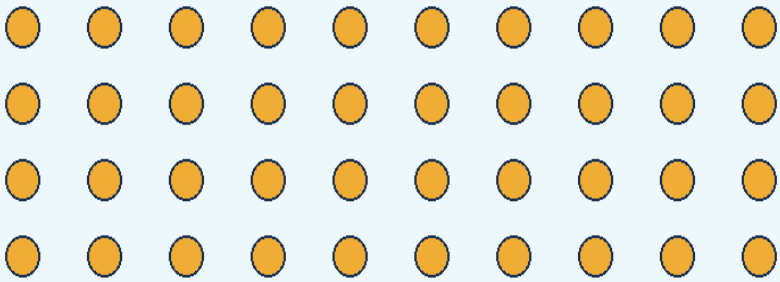
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Division facts to 81

LEARNING GOAL Division facts to 81

MATHS MODEL • DIVISION FACTS



4 groups of 10 = 40

LEARN THE IDEA

Division undoes multiplication: if 7×8=56, then 56÷7=8 and 56÷8=7.

WORKED EXAMPLE 1

Calculate 32 ÷ 8.

Answer: 4

8 × 4 = 32.

WORKED EXAMPLE 2

Calculate 72 ÷ 8.

Answer: 9

8 × 9 = 72.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Division facts to 81

PLAY · LEARN THE STEPS

1. **Calculate 16 ÷ 4.**
   A) 4 B) 6 C) 12 D) 5
2. **Calculate 24 ÷ 8.**
   A) 4 B) 3 C) 8 D) 16
3. **Calculate 36 ÷ 4.**
   A) 32 B) 9 C) 4 D) 10
4. **Calculate 21 ÷ 3.**
   A) 3 B) 8 C) 7 D) 18
5. **Calculate 15 ÷ 5.**
   A) 3 B) 5 C) 4 D) 10

CHALLENGE · DIFFERENT EXAMPLES

1. **A club at a science exhibition shares 21 pencils equally among 7 teams. How many per team?**
   A) 4 B) 7 C) 14 D) 3
2. **A club during a reading competition shares 20 pencils equally among 4 teams. How many per team?**
   A) 5 B) 6 C) 4 D) 16
3. **A club for a school fundraiser shares 10 pencils equally among 2 teams. How many per team?**
   A) 8 B) 2 C) 5 D) 6
4. **A club at a farm shares 35 pencils equally among 5 teams. How many per team?**
   A) 5 B) 7 C) 30 D) 8
5. **A club during a sports day shares 20 pencils equally among 4 teams. How many per team?**
   A) 6 B) 16 C) 5 D) 4

Division facts to 81

CHALLENGE · APPLY WHAT YOU KNOW

6. **A club in a craft workshop shares 12 pencils equally among 3 teams. How many per team?**
   A) 4 B) 3 C) 5 D) 9

7. **A club at a school fair shares 36 pencils equally among 4 teams. How many per team?**
   A) 4 B) 9 C) 32 D) 10

8. **A club in the class garden shares 15 pencils equally among 3 teams. How many per team?**
   A) 12 B) 6 C) 3 D) 5

9. **A club at a market stall shares 30 pencils equally among 5 teams. How many per team?**
   A) 25 B) 5 C) 6 D) 7

10. **A club during a library project shares 63 pencils equally among 7 teams. How many per team?**
    A) 56 B) 10 C) 7 D) 9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4	2. 3	3. 9
4. 7	5. 3	6. 3
7. 5	8. 5	9. 7
10. 5	11. 4	12. 9
13. 5	14. 6	15. 9

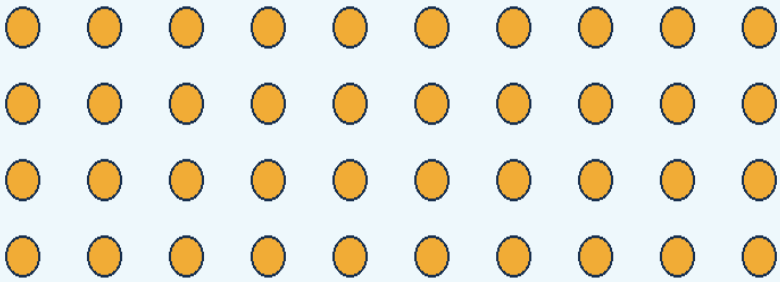
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Dividing three-digit numbers

LEARNING GOAL Dividing three-digit numbers

MATHS MODEL • LONG DIVISION



4 groups of 10 = 40

LEARN THE IDEA

Divide hundreds, then tens, then ones. check every answer by multiplication.

WORKED EXAMPLE 1

Calculate 186 ÷ 6.

Answer: 31

6 × 31 = 186.

WORKED EXAMPLE 2

Calculate 560 ÷ 8.

Answer: 70

8 × 70 = 560.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Dividing three-digit numbers

PLAY · LEARN THE STEPS

1. **Calculate 336 ÷ 6.**
   A) 6 B) 57 C) 56 D) 330
2. **Calculate 180 ÷ 5.**
   A) 37 B) 36 C) 175 D) 5
3. **Calculate 165 ÷ 5.**
   A) 34 B) 5 C) 160 D) 33
4. **Calculate 440 ÷ 8.**
   A) 55 B) 8 C) 432 D) 56
5. **Calculate 310 ÷ 5.**
   A) 5 B) 63 C) 305 D) 62

CHALLENGE · DIFFERENT EXAMPLES

1. **A club during a reading competition shares 531 pencils equally among 9 teams. How many per team?**
   A) 60 B) 59 C) 9 D) 522
2. **A club for a school fundraiser shares 423 pencils equally among 9 teams. How many per team?**
   A) 47 B) 9 C) 48 D) 414
3. **A club at a farm shares 285 pencils equally among 5 teams. How many per team?**
   A) 57 B) 58 C) 5 D) 280
4. **A club during a sports day shares 312 pencils equally among 6 teams. How many per team?**
   A) 6 B) 52 C) 53 D) 306
5. **A club in a craft workshop shares 225 pencils equally among 3 teams. How many per team?**
   A) 75 B) 222 C) 76 D) 3

Dividing three-digit numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **A club at a school fair shares 156 pencils equally among 4 teams. How many per team?**
   A) 40 B) 39 C) 4 D) 152

7. **A club in the class garden shares 228 pencils equally among 3 teams. How many per team?**
   A) 3 B) 76 C) 77 D) 225

8. **A club at a market stall shares 308 pencils equally among 4 teams. How many per team?**
   A) 4 B) 77 C) 304 D) 78

9. **A club during a library project shares 518 pencils equally among 7 teams. How many per team?**
   A) 74 B) 7 C) 75 D) 511

10. **A club on a football field shares 98 pencils equally among 2 teams. How many per team?**
    A) 49 B) 2 C) 50 D) 96

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 56	2. 36	3. 33
4. 55	5. 62	6. 59
7. 47	8. 57	9. 52
10. 75	11. 39	12. 76
13. 77	14. 74	15. 49

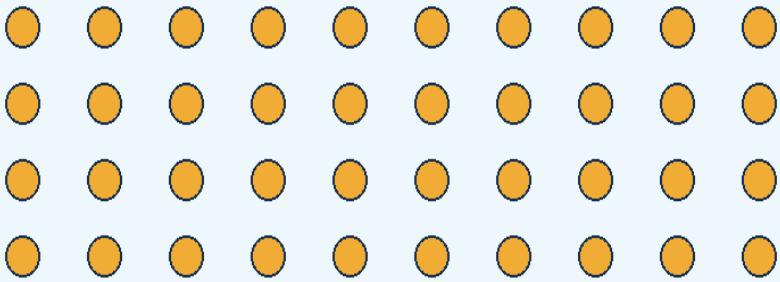
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multi-step word problems

LEARNING GOAL Multi-step word problems

MATHS MODEL • WORD PROBLEMS



4 groups of 10 = 40

LEARN THE IDEA

Read the question carefully, list the needed quantities, choose operations in order, then check units.

WORKED EXAMPLE 1

First add 39 and 33, then multiply the result by 2.

Answer: 144

(39+33) × 2 = 144.

WORKED EXAMPLE 2

First add 40 and 34, then multiply the result by 3.

Answer: 222

(40+34) × 3 = 222.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multi-step word problems

PLAY · LEARN THE STEPS

1. **First add 14 and 39, then multiply the result by 2.**
   A) 55 B) 53 C) 106 D) 108
2. **First add 39 and 26, then multiply the result by 4.**
   A) 264 B) 65 C) 260 D) 69
3. **First add 23 and 28, then multiply the result by 4.**
   A) 55 B) 51 C) 204 D) 208
4. **First add 31 and 15, then multiply the result by 5.**
   A) 51 B) 230 C) 46 D) 235
5. **First add 33 and 12, then multiply the result by 2.**
   A) 47 B) 45 C) 92 D) 90

CHALLENGE · DIFFERENT EXAMPLES

1. **A classroom for a school fundraiser has 35 blue counters and 26 red counters in each box. There are 3 boxes. How many counters altogether?**
   A) 186 B) 64 C) 183 D) 61
2. **A classroom at a farm has 16 blue counters and 33 red counters in each box. There are 4 boxes. How many counters altogether?**
   A) 196 B) 200 C) 53 D) 49
3. **A classroom during a sports day has 33 blue counters and 23 red counters in each box. There are 3 boxes. How many counters altogether?**
   A) 171 B) 56 C) 168 D) 59
4. **A classroom in a craft workshop has 22 blue counters and 12 red counters in each box. There are 5 boxes. How many counters altogether?**
   A) 34 B) 175 C) 170 D) 39
5. **A classroom at a school fair has 34 blue counters and 49 red counters in each box. There are 2 boxes. How many counters altogether?**
   A) 168 B) 85 C) 83 D) 166

Multi-step word problems

CHALLENGE · APPLY WHAT YOU KNOW

6 **A classroom in the class garden has 31 blue counters and 32 red counters in each box. There are 2 boxes. How many counters altogether?**

A) 128 B) 65 C) 63 D) 126

7 **A classroom at a market stall has 35 blue counters and 47 red counters in each box. There are 2 boxes. How many counters altogether?**

A) 164 B) 82 C) 84 D) 166

8 **A classroom during a library project has 16 blue counters and 18 red counters in each box. There are 4 boxes. How many counters altogether?**

A) 38 B) 34 C) 136 D) 140

9 **A classroom on a football field has 28 blue counters and 44 red counters in each box. There are 4 boxes. How many counters altogether?**

A) 72 B) 76 C) 288 D) 292

10 **A classroom in a recycling club has 36 blue counters and 40 red counters in each box. There are 5 boxes. How many counters altogether?**

A) 380 B) 385 C) 76 D) 81

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 106	2. 260	3. 204
4. 230	5. 90	6. 183
7. 196	8. 168	9. 170
10. 166	11. 126	12. 164
13. 136	14. 288	15. 380

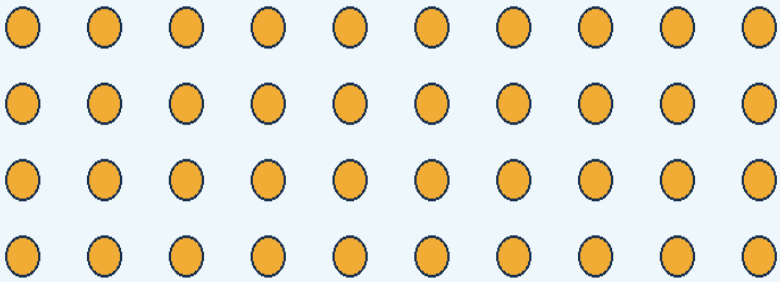
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting integers

LEARNING GOAL Adding and subtracting integers

MATHS MODEL • INTEGER



4 groups of 10 = 40

LEARN THE IDEA

Integers include positive and negative numbers. on the number line right adds and left subtracts.

WORKED EXAMPLE 1

Calculate -15 + (8).

Answer: -7

On the number line, start at -15 and move +8.

WORKED EXAMPLE 2

Calculate -9 + (12).

Answer: 3

On the number line, start at -9 and move +12.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Adding and subtracting integers

PLAY · LEARN THE STEPS

1. **Calculate 10 + (3).**
   A) 7 B) 15 C) 11 D) 13
2. **Calculate -8 + (5).**
   A) -5 B) -1 C) -3 D) -13
3. **Calculate -3 + (6).**
   A) 3 B) 5 C) 1 D) -9
4. **Calculate -9 + (9).**
   A) 2 B) -2 C) 0 D) -18
5. **Calculate -6 + (9).**
   A) -15 B) 5 C) 1 D) 3

CHALLENGE · DIFFERENT EXAMPLES

1. **The thermometer at a farm reads -14°C and changes by +12°C. What does it read afterward?**
   A) 0 B) -4 C) -26 D) -2
2. **The thermometer during a sports day reads 6°C and changes by +2°C. What does it read afterward?**
   A) 8 B) 4 C) 10 D) 6
3. **The thermometer in a craft workshop reads 8°C and changes by -10°C. What does it read afterward?**
   A) -2 B) 18 C) 0 D) -4
4. **The thermometer at a school fair reads 2°C and changes by -10°C. What does it read afterward?**
   A) -10 B) -6 C) 12 D) -8
5. **The thermometer in the class garden reads 9°C and changes by +8°C. What does it read afterward?**
   A) 15 B) 17 C) 1 D) 19

Adding and subtracting integers

CHALLENGE · APPLY WHAT YOU KNOW

6. **The thermometer at a market stall reads 6°C and changes by -9°C. What does it read afterward?**

   A) -3 B) -1 C) 15 D) -5

7. **The thermometer during a library project reads 5°C and changes by -11°C. What does it read afterward?**

   A) -8 B) -6 C) -4 D) 16

8. **The thermometer on a football field reads 13°C and changes by +5°C. What does it read afterward?**

   A) 8 B) 18 C) 20 D) 16

9. **The thermometer in a recycling club reads 9°C and changes by +11°C. What does it read afterward?**

   A) 20 B) -2 C) 18 D) 22

10. **The thermometer at a science exhibition reads 15°C and changes by +0°C. What does it read afterward?**

    A) 15 B) 13 C) 16 D) 17

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 13	2. -3	3. 3
4. 0	5. 3	6. -2
7. 8	8. -2	9. -8
10. 17	11. -3	12. -6
13. 18	14. 20	15. 15

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding equivalent fractions

LEARNING GOAL Finding equivalent fractions

MATHS MODEL • EQUIVALENT



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Multiply the numerator and denominator by the same nonzero number to make an equivalent fraction.

WORKED EXAMPLE 1

Complete: 1/5 = ?/15.

Answer: 3

Multiply numerator and denominator by 3: 3/15.

WORKED EXAMPLE 2

Complete: 5/8 = ?/40.

Answer: 25

Multiply numerator and denominator by 5: 25/40.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Finding equivalent fractions

PLAY · LEARN THE STEPS

1. **Complete: 1/5 = ?/10.**
   A) 1 B) 2 C) 3 D) 4
2. **Complete: 5/6 = ?/12.**
   A) 7 B) 10 C) 5 D) 11
3. **Complete: 2/3 = ?/9.**
   A) 6 B) 7 C) 5 D) 2
4. **Complete: 2/3 = ?/15.**
   A) 11 B) 7 C) 10 D) 2
5. **Complete: 1/4 = ?/12.**
   A) 3 B) 4 C) 5 D) 1

CHALLENGE · DIFFERENT EXAMPLES

1. **A recipe fraction during a sports day is 3/10. What is an equivalent fraction with denominator 30?**
   A) 6 B) 9 C) 10 D) 3
2. **A recipe fraction in a craft workshop is 3/5. What is an equivalent fraction with denominator 10?**
   A) 7 B) 6 C) 5 D) 3
3. **A recipe fraction at a school fair is 9/10. What is an equivalent fraction with denominator 40?**
   A) 36 B) 37 C) 9 D) 13
4. **A recipe fraction in the class garden is 2/3. What is an equivalent fraction with denominator 9?**
   A) 2 B) 5 C) 7 D) 6
5. **A recipe fraction at a market stall is 2/4. What is an equivalent fraction with denominator 12?**
   A) 6 B) 5 C) 2 D) 7

Finding equivalent fractions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A recipe fraction during a library project is 1/5. What is an equivalent fraction with denominator 10?**

A) 2 B) 3 C) 4 D) 1

7 **A recipe fraction on a football field is 1/4. What is an equivalent fraction with denominator 8?**

A) 2 B) 3 C) 1 D) 4

8 **A recipe fraction in a recycling club is 5/6. What is an equivalent fraction with denominator 18?**

A) 8 B) 5 C) 16 D) 15

9 **A recipe fraction at a science exhibition is 2/6. What is an equivalent fraction with denominator 30?**

A) 11 B) 7 C) 2 D) 10

10 **A recipe fraction during a reading competition is 7/10. What is an equivalent fraction with denominator 20?**

A) 9 B) 7 C) 15 D) 14

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 2	2. 10	3. 6
4. 10	5. 3	6. 9
7. 6	8. 36	9. 6
10. 6	11. 2	12. 2
13. 15	14. 10	15. 14

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing and ordering fractions

LEARNING GOAL Comparing and ordering fractions

MATHS MODEL • FRACTION COMPARE



6 / 10 = 0.6 = 60%

LEARN THE IDEA

For equal denominators, compare numerators. if denominators differ, first make equivalent fractions.

WORKED EXAMPLE 1

Compare 1/8 __ 5/8.

Answer: <

Both have denominator 8; compare 1 and 5.

WORKED EXAMPLE 2

Compare 2/4 __ 3/4.

Answer: <

Both have denominator 4; compare 2 and 3.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Comparing and ordering fractions

PLAY · LEARN THE STEPS

1. **Compare 4/12 __ 9/12.**
   A) none of these B) < C) = D) >
2. **Compare 3/5 __ 2/5.**
   A) < B) = C) > D) none of these
3. **Compare 4/6 __ 1/6.**
   A) < B) > C) none of these D) =
4. **Compare 3/4 __ 1/4.**
   A) < B) > C) none of these D) =
5. **Compare 6/8 __ 1/8.**
   A) none of these B) > C) < D) =

CHALLENGE · DIFFERENT EXAMPLES

1. **Two teams in a craft workshop completed 7/10 and 3/10 of their tasks. Choose the sign for 7/10 __ 3/10.**
   A) > B) none of these C) < D) =
2. **Two teams at a school fair completed 7/10 and 4/10 of their tasks. Choose the sign for 7/10 __ 4/10.**
   A) = B) none of these C) < D) >
3. **Two teams in the class garden completed 2/10 and 1/10 of their tasks. Choose the sign for 2/10 __ 1/10.**
   A) < B) = C) none of these D) >
4. **Two teams at a market stall completed 2/6 and 4/6 of their tasks. Choose the sign for 2/6 __ 4/6.**
   A) none of these B) > C) < D) =
5. **Two teams during a library project completed 10/12 and 4/12 of their tasks. Choose the sign for 10/12 __ 4/12.**
   A) > B) = C) < D) none of these

Comparing and ordering fractions

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two teams on a football field completed 3/5 and 1/5 of their tasks. Choose the sign for 3/5 __ 1/5.**

   A) < B) none of these C) = D) >

7. **Two teams in a recycling club completed 2/4 and 1/4 of their tasks. Choose the sign for 2/4 __ 1/4.**

   A) = B) < C) > D) none of these

8. **Two teams at a science exhibition completed 4/5 and 2/5 of their tasks. Choose the sign for 4/5 __ 2/5.**

   A) = B) none of these C) > D) <

9. **Two teams during a reading competition completed 5/6 and 4/6 of their tasks. Choose the sign for 5/6 __ 4/6.**

   A) none of these B) > C) = D) <

10. **Two teams for a school fundraiser completed 2/4 and 3/4 of their tasks. Choose the sign for 2/4 __ 3/4.**

    A) none of these B) = C) < D) >

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. <	2. >	3. >
4. >	5. >	6. >
7. >	8. >	9. <
10. >	11. >	12. >
13. >	14. >	15. <

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Making common denominators

LEARNING GOAL Making common denominators

MATHS MODEL • COMMON DENOM



6 / 10 = 0.6 = 60%

LEARN THE IDEA

A common denominator names equal-sized parts. multiply numerator and denominator together by the same scale.

WORKED EXAMPLE 1

Complete: 5/6 = ?/24.

Answer: 20

Multiply numerator and denominator by 4: 20/24.

WORKED EXAMPLE 2

Complete: 1/3 = ?/12.

Answer: 4

Multiply numerator and denominator by 4: 4/12.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Making common denominators

PLAY · LEARN THE STEPS

1. **Complete: 1/4 = ?/20.**
   A) 7 B) 5 C) 6 D) 1
2. **Complete: 7/10 = ?/20.**
   A) 15 B) 14 C) 7 D) 9
3. **Complete: 3/8 = ?/24.**
   A) 6 B) 10 C) 3 D) 9
4. **Complete: 1/3 = ?/6.**
   A) 4 B) 2 C) 1 D) 3
5. **Complete: 6/8 = ?/40.**
   A) 11 B) 31 C) 30 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A recipe fraction at a school fair is 1/6. What is an equivalent fraction with denominator 18?**
   A) 3 B) 1 C) 4 D) 5
2. **A recipe fraction in the class garden is 3/6. What is an equivalent fraction with denominator 30?**
   A) 15 B) 8 C) 3 D) 16
3. **A recipe fraction at a market stall is 1/6. What is an equivalent fraction with denominator 30?**
   A) 5 B) 1 C) 7 D) 6
4. **A recipe fraction during a library project is 1/8. What is an equivalent fraction with denominator 16?**
   A) 2 B) 4 C) 3 D) 1
5. **A recipe fraction on a football field is 4/8. What is an equivalent fraction with denominator 24?**
   A) 4 B) 12 C) 13 D) 7

Making common denominators

CHALLENGE · APPLY WHAT YOU KNOW

6 **A recipe fraction in a recycling club is 4/5. What is an equivalent fraction with denominator 25?**

A) 9 B) 4 C) 20 D) 21

7 **A recipe fraction at a science exhibition is 1/3. What is an equivalent fraction with denominator 15?**

A) 6 B) 1 C) 5 D) 7

8 **A recipe fraction during a reading competition is 3/5. What is an equivalent fraction with denominator 25?**

A) 3 B) 8 C) 16 D) 15

9 **A recipe fraction for a school fundraiser is 2/6. What is an equivalent fraction with denominator 18?**

A) 2 B) 6 C) 7 D) 5

10 **A recipe fraction at a farm is 3/4. What is an equivalent fraction with denominator 20?**

A) 3 B) 16 C) 8 D) 15

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5	2. 14	3. 9
4. 2	5. 30	6. 3
7. 15	8. 5	9. 2
10. 12	11. 20	12. 5
13. 15	14. 6	15. 15

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting fractions

LEARNING GOAL Adding and subtracting fractions

MATHS MODEL • FRACTION ADD



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Add or subtract numerators only after making denominators equal. simplify the final fraction.

WORKED EXAMPLE 1

Calculate 5/12 − 4/12.

Answer: 1/12

Work on the numerators, then simplify: 1/12.

WORKED EXAMPLE 2

Calculate 4/10 + 4/10.

Answer: 4/5

Work on the numerators, then simplify: 4/5.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Adding and subtracting fractions

PLAY · LEARN THE STEPS

1. **Calculate 5/8 − 3/8.**
   A) 3/8 B) 9/8 C) 1/4 D) 15/8
2. **Calculate 2/4 + 1/4.**
   A) 1/1 B) 4/4 C) 1/2 D) 3/4
3. **Calculate 3/8 − 1/8.**
   A) 1/4 B) 3/8 C) 2/4 D) 5/8
4. **Calculate 1/4 + 2/4.**
   A) 1/2 B) 4/4 C) 1/1 D) 3/4
5. **Calculate 4/8 − 2/8.**
   A) 7/8 B) 1/4 C) 3/8 D) 1/1

CHALLENGE · DIFFERENT EXAMPLES

1. **A measuring tape in the class garden starts at 5/8 metre and gains 2/8 metre. What length results?**
   A) 1/1 B) 7/8 C) 1/2 D) 5/4
2. **A measuring tape at a market stall starts at 3/8 metre and loses 2/8 metre. What length results?**
   A) 1/8 B) 1/4 C) 2/8 D) 3/4
3. **A measuring tape during a library project starts at 3/5 metre and gains 1/5 metre. What length results?**
   A) 3/5 B) 4/5 C) 5/5 D) 1/1
4. **A measuring tape on a football field starts at 6/10 metre and loses 3/10 metre. What length results?**
   A) 2/5 B) 1/1 C) 3/10 D) 9/5
5. **A measuring tape in a recycling club starts at 3/6 metre and gains 1/6 metre. What length results?**
   A) 3/3 B) 5/6 C) 1/2 D) 2/3

Adding and subtracting fractions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A measuring tape at a science exhibition starts at 4/12 metre and loses 1/12 metre. What length results?**

A) 1/2 B) 1/3 C) 2/4 D) 1/4

7 **A measuring tape during a reading competition starts at 2/10 metre and gains 6/10 metre. What length results?**

A) 6/5 B) 1/2 C) 9/10 D) 4/5

8 **A measuring tape for a school fundraiser starts at 4/6 metre and loses 2/6 metre. What length results?**

A) 4/3 B) 7/6 C) 1/2 D) 1/3

9 **A measuring tape at a farm starts at 2/10 metre and gains 2/10 metre. What length results?**

A) 1/10 B) 3/5 C) 1/2 D) 2/5

10 **A measuring tape during a sports day starts at 2/5 metre and loses 2/5 metre. What length results?**

A) 1/5 B) 1/1 C) 0/1 D) 4/5

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 1/4	2. 3/4	3. 1/4
4. 3/4	5. 1/4	6. 7/8
7. 1/8	8. 4/5	9. 3/10
10. 2/3	11. 1/4	12. 4/5
13. 1/3	14. 2/5	15. 0/1

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplying a whole number by a fraction

LEARNING GOAL Multiplying a whole number by a fraction

MATHS MODEL • FRACTION WHOLE



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Multiplying a fraction by a whole number means several equal fractional groups; use strips to model.

WORKED EXAMPLE 1

Calculate 3/8 of 48.

Answer: 18

One 8th of 48 is 6; multiply by 3 to get 18.

WORKED EXAMPLE 2

Calculate 1/5 of 35.

Answer: 7

One 5th of 35 is 7; multiply by 1 to get 7.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Multiplying a whole number by a fraction

PLAY · LEARN THE STEPS

1. **Calculate 5/8 of 48.**
   A) 30 B) 6 C) 35 D) 48
2. **Calculate 3/8 of 72.**
   A) 30 B) 9 C) 27 D) 72
3. **Calculate 1/6 of 30.**
   A) 6 B) 30 C) 7 D) 5
4. **Calculate 1/2 of 16.**
   A) 8 B) 16 C) 10 D) 9
5. **Calculate 4/6 of 36.**
   A) 6 B) 28 C) 24 D) 36

CHALLENGE · DIFFERENT EXAMPLES

1. **6/8 of 32 reading cards are used at a market stall. How many cards is that?**
   A) 24 B) 32 C) 30 D) 4
2. **2/6 of 48 reading cards are used during a library project. How many cards is that?**
   A) 18 B) 16 C) 48 D) 8
3. **1/4 of 8 reading cards are used on a football field. How many cards is that?**
   A) 2 B) 4 C) 3 D) 8
4. **3/8 of 40 reading cards are used in a recycling club. How many cards is that?**
   A) 5 B) 40 C) 15 D) 18
5. **2/6 of 54 reading cards are used at a science exhibition. How many cards is that?**
   A) 54 B) 9 C) 20 D) 18

Multiplying a whole number by a fraction

CHALLENGE · APPLY WHAT YOU KNOW

6. **7/8 of 48 reading cards are used during a reading competition. How many cards is that?**
   A) 6 B) 48 C) 49 D) 42

7. **1/4 of 36 reading cards are used for a school fundraiser. How many cards is that?**
   A) 11 B) 36 C) 9 D) 10

8. **3/8 of 64 reading cards are used at a farm. How many cards is that?**
   A) 27 B) 24 C) 64 D) 8

9. **2/4 of 8 reading cards are used during a sports day. How many cards is that?**
   A) 2 B) 6 C) 4 D) 8

10. **3/6 of 30 reading cards are used in a craft workshop. How many cards is that?**
    A) 18 B) 5 C) 15 D) 30

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 30	2. 27	3. 5
4. 8	5. 24	6. 24
7. 16	8. 2	9. 15
10. 18	11. 42	12. 9
13. 24	14. 4	15. 15

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding a fraction of a number

LEARNING GOAL Finding a fraction of a number

MATHS MODEL • OF FRACTION



6 / 10 = 0.6 = 60%

LEARN THE IDEA

To find a fraction of a quantity, divide into denominator-sized groups, then take the numerator number of groups.

WORKED EXAMPLE 1

Calculate 2/3 of 6.

Answer: 4

One 3th of 6 is 2; multiply by 2 to get 4.

WORKED EXAMPLE 2

Calculate 1/3 of 24.

Answer: 8

One 3th of 24 is 8; multiply by 1 to get 8.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Finding a fraction of a number

PLAY · LEARN THE STEPS

1. **Calculate 6/8 of 24.**
   A) 24 B) 3 C) 18 D) 19
2. **Calculate 4/8 of 56.**
   A) 7 B) 56 C) 32 D) 28
3. **Calculate 4/5 of 25.**
   A) 20 B) 24 C) 25 D) 5
4. **Calculate 5/6 of 12.**
   A) 12 B) 10 C) 2 D) 15
5. **Calculate 3/5 of 20.**
   A) 20 B) 12 C) 15 D) 4

CHALLENGE · DIFFERENT EXAMPLES

1. **2/3 of 24 reading cards are used during a library project. How many cards is that?**
   A) 24 B) 18 C) 16 D) 8
2. **1/2 of 4 reading cards are used on a football field. How many cards is that?**
   A) 2 B) 7 C) 3 D) 4
3. **1/4 of 16 reading cards are used in a recycling club. How many cards is that?**
   A) 16 B) 5 C) 6 D) 4
4. **2/6 of 36 reading cards are used at a science exhibition. How many cards is that?**
   A) 6 B) 36 C) 14 D) 12
5. **2/3 of 9 reading cards are used during a reading competition. How many cards is that?**
   A) 9 B) 8 C) 6 D) 3

Finding a fraction of a number

CHALLENGE · APPLY WHAT YOU KNOW

6. **1/3 of 21 reading cards are used for a school fundraiser. How many cards is that?**
A) 9 B) 21 C) 7 D) 8

7. **1/3 of 24 reading cards are used at a farm. How many cards is that?**
A) 8 B) 10 C) 24 D) 9

8. **4/8 of 56 reading cards are used during a sports day. How many cards is that?**
A) 7 B) 32 C) 28 D) 56

9. **1/3 of 6 reading cards are used in a craft workshop. How many cards is that?**
A) 2 B) 6 C) 4 D) 3

10. **2/4 of 32 reading cards are used at a school fair. How many cards is that?**
A) 8 B) 16 C) 18 D) 32

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 18	2. 28	3. 20
4. 10	5. 12	6. 16
7. 2	8. 4	9. 12
10. 6	11. 7	12. 8
13. 28	14. 2	15. 16

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Modelling tenths, hundredths and thousandths

LEARNING GOAL Modelling tenths, hundredths and thousandths

MATHS MODEL • DECIMAL MODEL



6 / 10 = 0.6 = 60%

LEARN THE IDEA

The first decimal place is tenths, next is hundredths and next is thousandths.

WORKED EXAMPLE 1

In 0.801, which digit occupies the hundredths place?

Answer: 0

0.801 has digit 0 in its hundredths place.

WORKED EXAMPLE 2

In 0.368, which digit occupies the tenths place?

Answer: 3

0.368 has digit 3 in its tenths place.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Modelling tenths, hundredths and thousandths

PLAY · LEARN THE STEPS

1. **In 0.929, which digit occupies the tenths place?**
   A) 9 B) 0 C) 4 D) 2
2. **In 0.314, which digit occupies the hundredths place?**
   A) 2 B) 1 C) 4 D) 6
3. **In 0.479, which digit occupies the thousandths place?**
   A) 2 B) 4 C) 9 D) 0
4. **In 0.442, which digit occupies the hundredths place?**
   A) 5 B) 9 C) 4 D) 7
5. **In 0.430, which digit occupies the tenths place?**
   A) 5 B) 4 C) 9 D) 7

CHALLENGE · DIFFERENT EXAMPLES

1. **A measurement on a football field is 0.211 m. Which digit is in the thousandths place?**
   A) 4 B) 2 C) 1 D) 6
2. **A measurement in a recycling club is 0.575 m. Which digit is in the hundredths place?**
   A) 7 B) 0 C) 2 D) 8
3. **A measurement at a science exhibition is 0.483 m. Which digit is in the hundredths place?**
   A) 8 B) 3 C) 9 D) 1
4. **A measurement during a reading competition is 0.581 m. Which digit is in the tenths place?**
   A) 8 B) 0 C) 6 D) 5
5. **A measurement for a school fundraiser is 0.732 m. Which digit is in the hundredths place?**
   A) 6 B) 3 C) 4 D) 8

Modelling tenths, hundredths and thousandths

CHALLENGE · APPLY WHAT YOU KNOW

6. **A measurement at a farm is 0.466 m. Which digit is in the hundredths place?**
A) 6 B) 1 C) 7 D) 9

7. **A measurement during a sports day is 0.368 m. Which digit is in the tenths place?**
A) 3 B) 6 C) 8 D) 4

8. **A measurement in a craft workshop is 0.800 m. Which digit is in the tenths place?**
A) 8 B) 9 C) 3 D) 1

9. **A measurement at a school fair is 0.790 m. Which digit is in the thousandths place?**
A) 3 B) 1 C) 0 D) 5

10. **A measurement in the class garden is 0.618 m. Which digit is in the tenths place?**
A) 6 B) 1 C) 7 D) 9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 9	2. 1	3. 9
4. 4	5. 4	6. 1
7. 7	8. 8	9. 5
10. 3	11. 6	12. 3
13. 8	14. 0	15. 6

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing decimals and fractions

LEARNING GOAL Comparing decimals and fractions

MATHS MODEL • DECIMAL COMPARE



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Line up decimal points and compare digits by place value, writing zeros if needed.

WORKED EXAMPLE 1

Compare 0.884 __ 0.822.

Answer: >

Compare thousandths: 0.884 is greater than 0.822.

WORKED EXAMPLE 2

Compare 0.581 __ 0.619.

Answer: <

Compare thousandths: 0.581 is less than 0.619.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Comparing decimals and fractions

PLAY · LEARN THE STEPS

1. **Compare 0.475 __ 0.640.**
   A) > B) none of these C) = D) <
2. **Compare 0.132 __ 0.558.**
   A) = B) none of these C) < D) >
3. **Compare 0.361 __ 0.516.**
   A) < B) = C) > D) none of these
4. **Compare 0.656 __ 0.417.**
   A) none of these B) = C) < D) >
5. **Compare 0.209 __ 0.344.**
   A) none of these B) = C) > D) <

CHALLENGE · DIFFERENT EXAMPLES

1. **Two measurements in a recycling club are 0.801 m and 0.875 m. Which sign completes 0.801 __ 0.875?**
   A) = B) none of these C) < D) >
2. **Two measurements at a science exhibition are 0.103 m and 0.745 m. Which sign completes 0.103 __ 0.745?**
   A) > B) = C) none of these D) <
3. **Two measurements during a reading competition are 0.121 m and 0.132 m. Which sign completes 0.121 __ 0.132?**
   A) > B) none of these C) = D) <
4. **Two measurements for a school fundraiser are 0.367 m and 0.769 m. Which sign completes 0.367 __ 0.769?**
   A) none of these B) < C) = D) >
5. **Two measurements at a farm are 0.776 m and 0.840 m. Which sign completes 0.776 __ 0.840?**
   A) < B) none of these C) = D) >

Comparing decimals and fractions

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two measurements during a sports day are 0.188 m and 0.661 m. Which sign completes 0.188 __ 0.661?**

A) none of these B) = C) > D) <

7 **Two measurements in a craft workshop are 0.471 m and 0.147 m. Which sign completes 0.471 __ 0.147?**

A) none of these B) < C) > D) =

8 **Two measurements at a school fair are 0.636 m and 0.869 m. Which sign completes 0.636 __ 0.869?**

A) > B) = C) none of these D) <

9 **Two measurements in the class garden are 0.466 m and 0.367 m. Which sign completes 0.466 __ 0.367?**

A) < B) = C) > D) none of these

10 **Two measurements at a market stall are 0.644 m and 0.665 m. Which sign completes 0.644 __ 0.665?**

A) none of these B) = C) < D) >

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. <	2. <	3. <
4. >	5. <	6. <
7. <	8. <	9. <
10. <	11. <	12. >
13. <	14. >	15. <

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Rounding decimals

LEARNING GOAL Rounding decimals

MATHS MODEL • DECIMAL ROUND



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Round a decimal by inspecting the digit just to the right of the last wanted place.

WORKED EXAMPLE 1

Round 0.378 to 1 decimal place(s).

Answer: 0.4

Look at the next decimal digit and round to 0.4.

WORKED EXAMPLE 2

Round 0.893 to 2 decimal place(s).

Answer: 0.89

Look at the next decimal digit and round to 0.89.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Rounding decimals

PLAY · LEARN THE STEPS

1. **Round 0.370 to 2 decimal place(s).**
   A) 0.36 B) 0.38 C) 0.37 D) 0.39
2. **Round 0.245 to 1 decimal place(s).**
   A) 0.2 B) 0.1 C) 0.4 D) 0.3
3. **Round 0.755 to 1 decimal place(s).**
   A) 0.9 B) 1.0 C) 0.7 D) 0.8
4. **Round 0.580 to 2 decimal place(s).**
   A) 0.60 B) 0.59 C) 0.58 D) 0.57
5. **Round 0.129 to 2 decimal place(s).**
   A) 0.15 B) 0.12 C) 0.13 D) 0.14

CHALLENGE · DIFFERENT EXAMPLES

1. **A measured length at a science exhibition is 0.717 metres. Round it to two decimal place(s).**
   A) 0.73 B) 0.71 C) 0.74 D) 0.72
2. **A measured length during a reading competition is 0.592 metres. Round it to one decimal place(s).**
   A) 0.7 B) 0.6 C) 0.8 D) 0.5
3. **A measured length for a school fundraiser is 0.770 metres. Round it to two decimal place(s).**
   A) 0.78 B) 0.76 C) 0.79 D) 0.77
4. **A measured length at a farm is 0.339 metres. Round it to one decimal place(s).**
   A) 0.5 B) 0.4 C) 0.2 D) 0.3
5. **A measured length during a sports day is 0.771 metres. Round it to one decimal place(s).**
   A) 1.0 B) 0.7 C) 0.9 D) 0.8

Rounding decimals

CHALLENGE · APPLY WHAT YOU KNOW

6. **A measured length in a craft workshop is 0.651 metres. Round it to one decimal place(s).**
   A) 0.7 B) 0.6 C) 0.8 D) 0.9

7. **A measured length at a school fair is 0.754 metres. Round it to one decimal place(s).**
   A) 0.7 B) 0.8 C) 1.0 D) 0.9

8. **A measured length in the class garden is 0.263 metres. Round it to one decimal place(s).**
   A) 0.5 B) 0.3 C) 0.4 D) 0.2

9. **A measured length at a market stall is 0.184 metres. Round it to one decimal place(s).**
   A) 0.3 B) 0.1 C) 0.2 D) 0.4

10. **A measured length during a library project is 0.483 metres. Round it to one decimal place(s).**
    A) 0.4 B) 0.5 C) 0.6 D) 0.7

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 0.37	2. 0.2	3. 0.8
4. 0.58	5. 0.13	6. 0.72
7. 0.6	8. 0.77	9. 0.3
10. 0.8	11. 0.7	12. 0.8
13. 0.3	14. 0.2	15. 0.5

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting decimals

LEARNING GOAL Adding and subtracting decimals

MATHS MODEL • DECIMAL ARITH



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Align decimal points when adding or subtracting; each column must have the same place value.

WORKED EXAMPLE 1

Calculate 2.63 − 1.37.

Answer: 1.26

Align the decimal points: 2.63 − 1.37 = 1.26.

WORKED EXAMPLE 2

Calculate 5.94 + 1.71.

Answer: 7.65

Align the decimal points: 5.94 + 1.71 = 7.65.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Adding and subtracting decimals

PLAY · LEARN THE STEPS

1. **Calculate 5.58 − 1.02.**
   A) 5.56 B) 4.56 C) 4.66 D) 4.36
2. **Calculate 3.72 + 1.43.**
   A) 5.25 B) 6.15 C) 5.15 D) 4.95
3. **Calculate 4.91 − 1.71.**
   A) 3.30 B) 3.20 C) 4.20 D) 3.00
4. **Calculate 1.35 + 1.89.**
   A) 3.24 B) 3.04 C) 4.24 D) 3.34
5. **Calculate 4.99 − 1.84.**
   A) 2.95 B) 4.15 C) 3.25 D) 3.15

CHALLENGE · DIFFERENT EXAMPLES

1. **Two measurements during a reading competition are 4.00 m and 1.15 m. What is the total?**
   A) 5.25 B) 5.15 C) 4.95 D) 6.15
2. **Two measurements for a school fundraiser are 5.28 m and 1.28 m. What is the difference?**
   A) 5.00 B) 3.80 C) 4.10 D) 4.00
3. **Two measurements at a farm are 3.43 m and 1.31 m. What is the total?**
   A) 5.74 B) 4.84 C) 4.74 D) 4.54
4. **Two measurements during a sports day are 7.52 m and 1.03 m. What is the difference?**
   A) 6.49 B) 6.29 C) 7.49 D) 6.59
5. **Two measurements in a craft workshop are 2.59 m and 1.75 m. What is the total?**
   A) 4.44 B) 5.34 C) 4.14 D) 4.34

Adding and subtracting decimals

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two measurements at a school fair are 7.27 m and 1.35 m. What is the difference?**
   A) 5.92 B) 6.02 C) 6.92 D) 5.72

7. **Two measurements in the class garden are 7.56 m and 1.71 m. What is the total?**
   A) 10.27 B) 9.37 C) 9.07 D) 9.27

8. **Two measurements at a market stall are 2.93 m and 1.17 m. What is the difference?**
   A) 2.76 B) 1.76 C) 1.86 D) 1.56

9. **Two measurements during a library project are 3.89 m and 1.69 m. What is the total?**
   A) 5.68 B) 5.38 C) 5.58 D) 6.58

10. **Two measurements on a football field are 5.58 m and 1.83 m. What is the difference?**
    A) 4.75 B) 3.85 C) 3.75 D) 3.55

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4.56	2. 5.15	3. 3.20
4. 3.24	5. 3.15	6. 5.15
7. 4.00	8. 4.74	9. 6.49
10. 4.34	11. 5.92	12. 9.27
13. 1.76	14. 5.58	15. 3.75

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplying with decimals

LEARNING GOAL Multiplying with decimals

MATHS MODEL • DECIMAL MULT



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Multiply the numbers as whole numbers then place the decimal point according to the number of decimal places.

WORKED EXAMPLE 1

Calculate 4.6 × 6.

Answer: 27.6

4.6 × 6 = 27.6.

WORKED EXAMPLE 2

Calculate 2.9 × 8.

Answer: 23.2

2.9 × 8 = 23.2.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multiplying with decimals

PLAY · LEARN THE STEPS

1. **Calculate 4.8 × 8.**
   A) 38.4 B) 30.4 C) 39.4 D) 46.4
2. **Calculate 2.9 × 9.**
   A) 17.1 B) 26.1 C) 27.1 D) 35.1
3. **Calculate 8.6 × 7.**
   A) 60.2 B) 53.2 C) 67.2 D) 61.2
4. **Calculate 2.4 × 5.**
   A) 7.0 B) 13.0 C) 12.0 D) 17.0
5. **Calculate 5.0 × 7.**
   A) 28.0 B) 42.0 C) 36.0 D) 35.0

CHALLENGE · DIFFERENT EXAMPLES

1. **A ribbon for a school fundraiser is 6.8 metres long. How many metres are in 7 equal ribbons?**
   A) 48.6 B) 54.6 C) 40.6 D) 47.6
2. **A ribbon at a farm is 2.8 metres long. How many metres are in 4 equal ribbons?**
   A) 15.2 B) 12.2 C) 11.2 D) 7.2
3. **A ribbon during a sports day is 1.9 metres long. How many metres are in 4 equal ribbons?**
   A) 7.6 B) 8.6 C) 11.6 D) 3.6
4. **A ribbon in a craft workshop is 3.0 metres long. How many metres are in 3 equal ribbons?**
   A) 10.0 B) 9.0 C) 12.0 D) 6.0
5. **A ribbon at a school fair is 2.3 metres long. How many metres are in 6 equal ribbons?**
   A) 13.8 B) 7.8 C) 14.8 D) 19.8

Multiplying with decimals

CHALLENGE · APPLY WHAT YOU KNOW

6 **A ribbon in the class garden is 7.9 metres long. How many metres are in 6 equal ribbons?**

A) 53.4 B) 41.4 C) 48.4 D) 47.4

7 **A ribbon at a market stall is 1.1 metres long. How many metres are in 4 equal ribbons?**

A) 0.4 B) 4.4 C) 5.4 D) 8.4

8 **A ribbon during a library project is 9.4 metres long. How many metres are in 4 equal ribbons?**

A) 37.6 B) 38.6 C) 41.6 D) 33.6

9 **A ribbon on a football field is 2.2 metres long. How many metres are in 7 equal ribbons?**

A) 8.4 B) 15.4 C) 16.4 D) 22.4

10 **A ribbon in a recycling club is 9.0 metres long. How many metres are in 7 equal ribbons?**

A) 70.0 B) 64.0 C) 56.0 D) 63.0

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 38.4	2. 26.1	3. 60.2
4. 12.0	5. 35.0	6. 47.6
7. 11.2	8. 7.6	9. 9.0
10. 13.8	11. 47.4	12. 4.4
13. 37.6	14. 15.4	15. 63.0

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Percentages of quantities

LEARNING GOAL Percentages of quantities

MATHS MODEL • PERCENT



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Percent means out of 100. multiply the whole amount by the fraction represented by the percentage.

WORKED EXAMPLE 1

Find 75% of 80.

Answer: 60

75% = 75/100, so 80 × 75/100 = 60.

WORKED EXAMPLE 2

Find 50% of 80.

Answer: 40

50% = 50/100, so 80 × 50/100 = 40.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Percentages of quantities

PLAY · LEARN THE STEPS

1. **Find 20% of 40.**
   A) 20 B) 8 C) 32 D) 13
2. **Find 75% of 100.**
   A) 80 B) 76 C) 75 D) 25
3. **Find 50% of 20.**
   A) 50 B) 11 C) 15 D) 10
4. **Find 10% of 40.**
   A) 36 B) 4 C) 9 D) 10
5. **Find 25% of 80.**
   A) 60 B) 25 C) 21 D) 20

CHALLENGE · DIFFERENT EXAMPLES

1. **A group at a farm has 60 learners; 10% joined an activity. How many joined?**
   A) 11 B) 10 C) 6 D) 54
2. **A group during a sports day has 20 learners; 25% joined an activity. How many joined?**
   A) 25 B) 15 C) 10 D) 5
3. **A group in a craft workshop has 40 learners; 75% joined an activity. How many joined?**
   A) 35 B) 10 C) 30 D) 75
4. **A group at a school fair has 20 learners; 50% joined an activity. How many joined?**
   A) 10 B) 11 C) 15 D) 50
5. **A group in the class garden has 20 learners; 50% joined an activity. How many joined?**
   A) 50 B) 15 C) 11 D) 10

Percentages of quantities

CHALLENGE · APPLY WHAT YOU KNOW

6 **A group at a market stall has 100 learners; 75% joined an activity. How many joined?**

A) 80 B) 76 C) 75 D) 25

7 **A group during a library project has 40 learners; 75% joined an activity. How many joined?**

A) 30 B) 35 C) 10 D) 75

8 **A group on a football field has 60 learners; 10% joined an activity. How many joined?**

A) 6 B) 54 C) 10 D) 11

9 **A group in a recycling club has 80 learners; 75% joined an activity. How many joined?**

A) 60 B) 65 C) 75 D) 20

10 **A group at a science exhibition has 60 learners; 50% joined an activity. How many joined?**

A) 31 B) 30 C) 35 D) 50

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 8	2. 75	3. 10
4. 4	5. 20	6. 6
7. 5	8. 30	9. 10
10. 10	11. 75	12. 30
13. 6	14. 60	15. 30

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Benchmark percentage facts

LEARNING GOAL Benchmark percentage facts

MATHS MODEL • BENCH PERCENT



6 / 10 = 0.6 = 60%

LEARN THE IDEA

Remember benchmarks: 50% is a half, 25% is a quarter, 10% is a tenth and 75% is three quarters.

WORKED EXAMPLE 1

Find 10% of 20.

Answer: 2

10% = 10/100, so 20 × 10/100 = 2.

WORKED EXAMPLE 2

Find 20% of 20.

Answer: 4

20% = 20/100, so 20 × 20/100 = 4.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Benchmark percentage facts

PLAY · LEARN THE STEPS

1. **Find 25% of 100.**
   A) 26 B) 30 C) 25 D) 75
2. **Find 50% of 100.**
   A) 52 B) 55 C) 51 D) 50
3. **Find 50% of 60.**
   A) 50 B) 30 C) 35 D) 31
4. **Find 25% of 80.**
   A) 20 B) 60 C) 21 D) 25
5. **Find 50% of 80.**
   A) 41 B) 40 C) 50 D) 45

CHALLENGE · DIFFERENT EXAMPLES

1. **For a total of 100 items during a sports day, what is 25%?**
   A) 25 B) 26 C) 75 D) 30
2. **For a total of 60 items in a craft workshop, what is 20%?**
   A) 48 B) 17 C) 12 D) 20
3. **For a total of 20 items at a school fair, what is 10%?**
   A) 2 B) 18 C) 7 D) 10
4. **For a total of 60 items in the class garden, what is 10%?**
   A) 54 B) 10 C) 6 D) 11
5. **For a total of 100 items at a market stall, what is 20%?**
   A) 20 B) 25 C) 80 D) 21

Benchmark percentage facts

CHALLENGE · APPLY WHAT YOU KNOW

6. **For a total of 40 items during a library project, what is 25%?**
   A) 25 B) 30 C) 15 D) 10

7. **For a total of 80 items on a football field, what is 75%?**
   A) 65 B) 60 C) 75 D) 20

8. **For a total of 60 items in a recycling club, what is 10%?**
   A) 11 B) 10 C) 6 D) 54

9. **For a total of 40 items at a science exhibition, what is 75%?**
   A) 10 B) 35 C) 30 D) 75

10. **For a total of 100 items during a reading competition, what is 25%?**
    A) 26 B) 75 C) 25 D) 30

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 25	2. 50	3. 30
4. 20	5. 40	6. 25
7. 12	8. 2	9. 6
10. 20	11. 10	12. 60
13. 6	14. 30	15. 25

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Percentages in everyday situations

LEARNING GOAL Percentages in everyday situations

MATHS MODEL • PERCENT CONTEXT



$$6 / 10 = 0.6 = 60\%$$

LEARN THE IDEA

Connect everyday discounts, participation and surveys to percentages; always identify the whole.

WORKED EXAMPLE 1

Find 25% of 20.

Answer: 5

25% = 25/100, so 20 × 25/100 = 5.

WORKED EXAMPLE 2

Find 75% of 40.

Answer: 30

75% = 75/100, so 40 × 75/100 = 30.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Percentages in everyday situations

PLAY · LEARN THE STEPS

1. **Find 10% of 80.**
   A) 8 B) 10 C) 13 D) 72
2. **Find 75% of 40.**
   A) 75 B) 30 C) 10 D) 35
3. **Find 50% of 20.**
   A) 50 B) 15 C) 10 D) 11
4. **Find 10% of 40.**
   A) 10 B) 36 C) 4 D) 9
5. **Find 75% of 100.**
   A) 75 B) 25 C) 80 D) 76

CHALLENGE · DIFFERENT EXAMPLES

1. **A group in a craft workshop has 60 learners; 10% joined an activity. How many joined?**
   A) 54 B) 11 C) 6 D) 10
2. **A group at a school fair has 100 learners; 50% joined an activity. How many joined?**
   A) 50 B) 55 C) 51 D) 52
3. **A group in the class garden has 20 learners; 10% joined an activity. How many joined?**
   A) 10 B) 18 C) 7 D) 2
4. **A group at a market stall has 40 learners; 25% joined an activity. How many joined?**
   A) 15 B) 25 C) 30 D) 10
5. **A group during a library project has 40 learners; 50% joined an activity. How many joined?**
   A) 50 B) 20 C) 25 D) 21

Percentages in everyday situations

CHALLENGE · APPLY WHAT YOU KNOW

6 **A group on a football field has 40 learners; 20% joined an activity. How many joined?**

A) 20 B) 32 C) 13 D) 8

7 **A group in a recycling club has 100 learners; 20% joined an activity. How many joined?**

A) 21 B) 20 C) 80 D) 25

8 **A group at a science exhibition has 60 learners; 10% joined an activity. How many joined?**

A) 11 B) 6 C) 54 D) 10

9 **A group during a reading competition has 80 learners; 75% joined an activity. How many joined?**

A) 75 B) 65 C) 60 D) 20

10 **A group for a school fundraiser has 40 learners; 75% joined an activity. How many joined?**

A) 10 B) 35 C) 75 D) 30

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 8	2. 30	3. 10
4. 4	5. 75	6. 6
7. 50	8. 2	9. 10
10. 20	11. 8	12. 20
13. 6	14. 60	15. 30

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Extending and describing patterns

LEARNING GOAL Extending and describing patterns

MATHS MODEL • PATTERNS



Describe the rule, then explain why it works.

LEARN THE IDEA

Find the operation connecting consecutive terms. use the same step to extend the sequence and justify it.

WORKED EXAMPLE 1

Continue: 5, 10, 15, ... What is term 7?

Answer: 35

Term 7 is 5+6×5 = 35.

WORKED EXAMPLE 2

Continue: 3, 9, 15, ... What is term 8?

Answer: 45

Term 8 is 3+7×6 = 45.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Extending and describing patterns

PLAY · LEARN THE STEPS

1. **Continue: 7, 17, 27, ... What is term 7?**
   A) 67 B) 57 C) 68 D) 77
2. **Continue: 20, 26, 32, ... What is term 5?**
   A) 45 B) 44 C) 38 D) 50
3. **Continue: 1, 11, 21, ... What is term 6?**
   A) 61 B) 51 C) 41 D) 52
4. **Continue: 23, 33, 43, ... What is term 5?**
   A) 53 B) 73 C) 63 D) 64
5. **Continue: 23, 25, 27, ... What is term 8?**
   A) 37 B) 35 C) 39 D) 38

CHALLENGE · DIFFERENT EXAMPLES

1. **The values 6, 9, 12 describe a pattern at a school fair. What is the value at step 6 (counting the first as step 1)?**
   A) 24 B) 22 C) 18 D) 21
2. **The values 22, 25, 28 describe a pattern in the class garden. What is the value at step 5 (counting the first as step 1)?**
   A) 31 B) 35 C) 34 D) 37
3. **The values 10, 16, 22 describe a pattern at a market stall. What is the value at step 5 (counting the first as step 1)?**
   A) 28 B) 34 C) 35 D) 40
4. **The values 19, 26, 33 describe a pattern during a library project. What is the value at step 8 (counting the first as step 1)?**
   A) 61 B) 75 C) 69 D) 68
5. **The values 7, 15, 23 describe a pattern on a football field. What is the value at step 5 (counting the first as step 1)?**
   A) 47 B) 31 C) 40 D) 39

Extending and describing patterns

CHALLENGE · APPLY WHAT YOU KNOW

6 **The values 20, 29, 38 describe a pattern in a recycling club. What is the value at step 7 (counting the first as step 1)?**

A) 65 B) 83 C) 74 D) 75

7 **The values 7, 15, 23 describe a pattern at a science exhibition. What is the value at step 5 (counting the first as step 1)?**

A) 47 B) 31 C) 39 D) 40

8 **The values 9, 19, 29 describe a pattern during a reading competition. What is the value at step 8 (counting the first as step 1)?**

A) 69 B) 79 C) 80 D) 89

9 **The values 19, 28, 37 describe a pattern for a school fundraiser. What is the value at step 5 (counting the first as step 1)?**

A) 56 B) 46 C) 64 D) 55

10 **The values 7, 9, 11 describe a pattern at a farm. What is the value at step 8 (counting the first as step 1)?**

A) 23 B) 22 C) 19 D) 21

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 67	2. 44	3. 51
4. 63	5. 37	6. 21
7. 34	8. 34	9. 68
10. 39	11. 74	12. 39
13. 79	14. 55	15. 21

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Predicting and drawing patterns

LEARNING GOAL Predicting and drawing patterns

MATHS MODEL • PATTERNS



Describe the rule, then explain why it works.

LEARN THE IDEA

Find the operation connecting consecutive terms. use the same step to extend the sequence and justify it.

WORKED EXAMPLE 1

Continue: 23, 28, 33, ... What is term 8?

Answer: 58

Term 8 is 23+7×5 = 58.

WORKED EXAMPLE 2

Continue: 1, 6, 11, ... What is term 7?

Answer: 31

Term 7 is 1+6×5 = 31.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Predicting and drawing patterns

PLAY · LEARN THE STEPS

1. **Continue: 19, 21, 23, ... What is term 8?**
   A) 35 B) 33 C) 31 D) 34
2. **Continue: 6, 16, 26, ... What is term 8?**
   A) 77 B) 76 C) 86 D) 66
3. **Continue: 20, 24, 28, ... What is term 7?**
   A) 40 B) 48 C) 44 D) 45
4. **Continue: 25, 33, 41, ... What is term 5?**
   A) 65 B) 57 C) 49 D) 58
5. **Continue: 14, 16, 18, ... What is term 6?**
   A) 26 B) 24 C) 25 D) 22

CHALLENGE · DIFFERENT EXAMPLES

1. **The values 2, 12, 22 describe a pattern in the class garden. What is the value at step 8 (counting the first as step 1)?**
   A) 73 B) 62 C) 82 D) 72
2. **The values 9, 12, 15 describe a pattern at a market stall. What is the value at step 8 (counting the first as step 1)?**
   A) 31 B) 33 C) 30 D) 27
3. **The values 8, 18, 28 describe a pattern during a library project. What is the value at step 7 (counting the first as step 1)?**
   A) 78 B) 68 C) 69 D) 58
4. **The values 16, 19, 22 describe a pattern on a football field. What is the value at step 5 (counting the first as step 1)?**
   A) 28 B) 29 C) 25 D) 31
5. **The values 14, 17, 20 describe a pattern in a recycling club. What is the value at step 6 (counting the first as step 1)?**
   A) 29 B) 30 C) 32 D) 26

Predicting and drawing patterns

CHALLENGE · APPLY WHAT YOU KNOW

6 **The values 19, 23, 27 describe a pattern at a science exhibition. What is the value at step 5 (counting the first as step 1)?**

A) 36 B) 35 C) 39 D) 31

7 **The values 14, 20, 26 describe a pattern during a reading competition. What is the value at step 7 (counting the first as step 1)?**

A) 50 B) 56 C) 51 D) 44

8 **The values 2, 9, 16 describe a pattern for a school fundraiser. What is the value at step 6 (counting the first as step 1)?**

A) 37 B) 44 C) 38 D) 30

9 **The values 21, 23, 25 describe a pattern at a farm. What is the value at step 5 (counting the first as step 1)?**

A) 27 B) 31 C) 30 D) 29

10 **The values 3, 6, 9 describe a pattern during a sports day. What is the value at step 7 (counting the first as step 1)?**

A) 18 B) 22 C) 24 D) 21

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 33	2. 76	3. 44
4. 57	5. 24	6. 72
7. 30	8. 68	9. 28
10. 29	11. 35	12. 50
13. 37	14. 29	15. 21

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Using patterns to solve problems

LEARNING GOAL Using patterns to solve problems

MATHS MODEL • PATTERNS



Describe the rule, then explain why it works.

LEARN THE IDEA

Find the operation connecting consecutive terms. use the same step to extend the sequence and justify it.

WORKED EXAMPLE 1

Continue: 24, 31, 38, ... What is term 7?

Answer: 66

Term 7 is 24+6×7 = 66.

WORKED EXAMPLE 2

Continue: 23, 28, 33, ... What is term 8?

Answer: 58

Term 8 is 23+7×5 = 58.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Using patterns to solve problems

PLAY · LEARN THE STEPS

1. **Continue: 9, 17, 25, ... What is term 5?**
   A) 49 B) 33 C) 42 D) 41
2. **Continue: 9, 18, 27, ... What is term 6?**
   A) 55 B) 54 C) 63 D) 45
3. **Continue: 7, 9, 11, ... What is term 7?**
   A) 17 B) 21 C) 20 D) 19
4. **Continue: 13, 20, 27, ... What is term 5?**
   A) 42 B) 34 C) 48 D) 41
5. **Continue: 11, 20, 29, ... What is term 7?**
   A) 56 B) 66 C) 65 D) 74

CHALLENGE · DIFFERENT EXAMPLES

1. **The values 22, 29, 36 describe a pattern at a market stall. What is the value at step 8 (counting the first as step 1)?**
   A) 64 B) 72 C) 78 D) 71
2. **The values 19, 27, 35 describe a pattern during a library project. What is the value at step 5 (counting the first as step 1)?**
   A) 43 B) 51 C) 59 D) 52
3. **The values 16, 18, 20 describe a pattern on a football field. What is the value at step 6 (counting the first as step 1)?**
   A) 28 B) 26 C) 24 D) 27
4. **The values 20, 23, 26 describe a pattern in a recycling club. What is the value at step 7 (counting the first as step 1)?**
   A) 39 B) 35 C) 38 D) 41
5. **The values 10, 17, 24 describe a pattern at a science exhibition. What is the value at step 6 (counting the first as step 1)?**
   A) 45 B) 46 C) 52 D) 38

Using patterns to solve problems

CHALLENGE · APPLY WHAT YOU KNOW

6 **The values 9, 16, 23 describe a pattern during a reading competition. What is the value at step 7 (counting the first as step 1)?**

A) 58 B) 51 C) 52 D) 44

7 **The values 11, 15, 19 describe a pattern for a school fundraiser. What is the value at step 6 (counting the first as step 1)?**

A) 27 B) 31 C) 32 D) 35

8 **The values 2, 5, 8 describe a pattern at a farm. What is the value at step 5 (counting the first as step 1)?**

A) 14 B) 15 C) 17 D) 11

9 **The values 23, 33, 43 describe a pattern during a sports day. What is the value at step 6 (counting the first as step 1)?**

A) 74 B) 63 C) 73 D) 83

10 **The values 9, 19, 29 describe a pattern in a craft workshop. What is the value at step 7 (counting the first as step 1)?**

A) 79 B) 59 C) 70 D) 69

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 41	2. 54	3. 19
4. 41	5. 65	6. 71
7. 51	8. 26	9. 38
10. 45	11. 51	12. 31
13. 14	14. 73	15. 69

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Writing a pattern rule

LEARNING GOAL Writing a pattern rule

MATHS MODEL • PATTERN RULE



Describe the rule, then explain why it works.

LEARN THE IDEA

A pattern rule expresses the starting amount and change using a letter such as n.

WORKED EXAMPLE 1

A rule starts at 22 and increases by 5 per step. Which expression describes it?

Answer: 22 + 5n

Initial value 22, then add 5 for each step n.

WORKED EXAMPLE 2

A rule starts at 12 and increases by 2 per step. Which expression describes it?

Answer: 12 + 2n

Initial value 12, then add 2 for each step n.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Writing a pattern rule

PLAY · LEARN THE STEPS

1. **A rule starts at 23 and increases by 4 per step. Which expression describes it?**
   A) 23 - 4n B) 23 + 4n C) 23n + 4 D) 23 + n + 4
2. **A rule starts at 24 and increases by 8 per step. Which expression describes it?**
   A) 24n + 8 B) 24 + 8n C) 24 - 8n D) 24 + n + 8
3. **A rule starts at 2 and increases by 4 per step. Which expression describes it?**
   A) 2 + n + 4 B) 2 + 4n C) 2n + 4 D) 2 - 4n
4. **A rule starts at 19 and increases by 10 per step. Which expression describes it?**
   A) 19 + n + 10 B) 19 + 10n C) 19n + 10 D) 19 - 10n
5. **A rule starts at 4 and increases by 10 per step. Which expression describes it?**
   A) 4n + 10 B) 4 + n + 10 C) 4 - 10n D) 4 + 10n

CHALLENGE · DIFFERENT EXAMPLES

1. **A price table during a library project starts at 19 and adds 2 for each new step. Which expression gives the value at step n?**
   A) 19 + 2n B) 19n + 2 C) 19 + n + 2 D) 19 - 2n
2. **A price table on a football field starts at 19 and adds 9 for each new step. Which expression gives the value at step n?**
   A) 19 + n + 9 B) 19 + 9n C) 19n + 9 D) 19 - 9n
3. **A price table in a recycling club starts at 22 and adds 4 for each new step. Which expression gives the value at step n?**
   A) 22 - 4n B) 22n + 4 C) 22 + 4n D) 22 + n + 4
4. **A price table at a science exhibition starts at 4 and adds 4 for each new step. Which expression gives the value at step n?**
   A) 4 - 4n B) 4 + 4n C) 4n + 4 D) 4 + n + 4
5. **A price table during a reading competition starts at 16 and adds 10 for each new step. Which expression gives the value at step n?**
   A) 16 + n + 10 B) 16n + 10 C) 16 + 10n D) 16 - 10n

Writing a pattern rule

CHALLENGE · APPLY WHAT YOU KNOW

6 **A price table for a school fundraiser starts at 21 and adds 2 for each new step. Which expression gives the value at step n?**

A) 21 + n + 2 B) 21 + 2n C) 21n + 2 D) 21 - 2n

7 **A price table at a farm starts at 25 and adds 10 for each new step. Which expression gives the value at step n?**

A) 25 + 10n B) 25n + 10 C) 25 - 10n D) 25 + n + 10

8 **A price table during a sports day starts at 15 and adds 7 for each new step. Which expression gives the value at step n?**

A) 15 + n + 7 B) 15n + 7 C) 15 - 7n D) 15 + 7n

9 **A price table in a craft workshop starts at 13 and adds 2 for each new step. Which expression gives the value at step n?**

A) 13n + 2 B) 13 - 2n C) 13 + n + 2 D) 13 + 2n

10 **A price table at a school fair starts at 7 and adds 4 for each new step. Which expression gives the value at step n?**

A) 7 + 4n B) 7 + n + 4 C) 7 - 4n D) 7n + 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 23 + 4n	2. 24 + 8n	3. 2 + 4n
4. 19 + 10n	5. 4 + 10n	6. 19 + 2n
7. 19 + 9n	8. 22 + 4n	9. 4 + 4n
10. 16 + 10n	11. 21 + 2n	12. 25 + 10n
13. 15 + 7n	14. 13 + 2n	15. 7 + 4n

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Building algebraic expressions

LEARNING GOAL Building algebraic expressions

MATHS MODEL • EXPRESSIONS



Describe the rule, then explain why it works.

LEARN THE IDEA

An algebraic expression uses symbols and numbers. substitute a value for the letter to evaluate it.

WORKED EXAMPLE 1

Evaluate 4x + 9 when x = 11.

Answer: 53

Substitute x=11: 4×11+9=53.

WORKED EXAMPLE 2

Evaluate 8x + 1 when x = 11.

Answer: 89

Substitute x=11: 8×11+1=89.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Building algebraic expressions

PLAY · LEARN THE STEPS

1. **Evaluate 4x + 3 when x = 12.**
   A) 55 B) 19 C) 51 D) 45
2. **Evaluate 2x + 1 when x = 9.**
   A) 19 B) 21 C) 17 D) 12
3. **Evaluate 7x + 4 when x = 10.**
   A) 81 B) 66 C) 21 D) 74
4. **Evaluate 3x + 10 when x = 11.**
   A) 43 B) 24 C) 23 D) 46
5. **Evaluate 7x + 3 when x = 8.**
   A) 18 B) 66 C) 59 D) 53

CHALLENGE · DIFFERENT EXAMPLES

1. **A fee on a football field is 2 cedis plus 2 cedis per item. For 7 items, what is the total?**
   A) 16 B) 18 C) 12 D) 11
2. **A fee in a recycling club is 6 cedis plus 8 cedis per item. For 6 items, what is the total?**
   A) 62 B) 54 C) 20 D) 42
3. **A fee at a science exhibition is 4 cedis plus 7 cedis per item. For 9 items, what is the total?**
   A) 59 B) 20 C) 74 D) 67
4. **A fee during a reading competition is 13 cedis plus 9 cedis per item. For 6 items, what is the total?**
   A) 76 B) 28 C) 67 D) 41
5. **A fee for a school fundraiser is 8 cedis plus 4 cedis per item. For 10 items, what is the total?**
   A) 32 B) 52 C) 22 D) 48

Building algebraic expressions

CHALLENGE · APPLY WHAT YOU KNOW

6. **A fee at a farm is 7 cedis plus 6 cedis per item. For 11 items, what is the total?**
A) 24 B) 59 C) 73 D) 79

7. **A fee during a sports day is 3 cedis plus 5 cedis per item. For 4 items, what is the total?**
A) 12 B) 28 C) 23 D) 17

8. **A fee in a craft workshop is 1 cedis plus 3 cedis per item. For 10 items, what is the total?**
A) 29 B) 31 C) 14 D) 34

9. **A fee at a school fair is 9 cedis plus 8 cedis per item. For 9 items, what is the total?**
A) 26 B) 89 C) 63 D) 81

10. **A fee in the class garden is 13 cedis plus 7 cedis per item. For 11 items, what is the total?**
A) 31 B) 64 C) 97 D) 90

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 51	2. 19	3. 74
4. 43	5. 59	6. 16
7. 54	8. 67	9. 67
10. 48	11. 73	12. 23
13. 31	14. 81	15. 90

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Representing and solving equations

LEARNING GOAL Representing and solving equations

MATHS MODEL • EQUATIONS



Describe the rule, then explain why it works.

LEARN THE IDEA

An equation says two quantities are equal. use inverse operations to solve for the unknown.

WORKED EXAMPLE 1

Solve x + 4 = 19.

Answer: 15

x = 19 − 4 = 15.

WORKED EXAMPLE 2

Solve x + 11 = 38.

Answer: 27

x = 38 − 11 = 27.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Representing and solving equations

PLAY · LEARN THE STEPS

1. **Solve x + 12 = 42.**
   A) 30 B) 42 C) 12 D) 32
2. **Solve x + 7 = 35.**
   A) 7 B) 28 C) 35 D) 30
3. **Solve x + 10 = 39.**
   A) 31 B) 39 C) 29 D) 10
4. **Solve x + 16 = 23.**
   A) 16 B) 23 C) 9 D) 7
5. **Solve x + 21 = 46.**
   A) 21 B) 25 C) 46 D) 27

CHALLENGE · DIFFERENT EXAMPLES

1. **A box in a recycling club has 24 stickers plus x more; altogether there are 40. What is x?**
   A) 16 B) 18 C) 24 D) 40
2. **A box at a science exhibition has 11 stickers plus x more; altogether there are 40. What is x?**
   A) 31 B) 29 C) 40 D) 11
3. **A box during a reading competition has 22 stickers plus x more; altogether there are 46. What is x?**
   A) 24 B) 22 C) 46 D) 26
4. **A box for a school fundraiser has 25 stickers plus x more; altogether there are 35. What is x?**
   A) 25 B) 10 C) 12 D) 35
5. **A box at a farm has 15 stickers plus x more; altogether there are 38. What is x?**
   A) 23 B) 15 C) 38 D) 25

Representing and solving equations

CHALLENGE · APPLY WHAT YOU KNOW

6 **A box during a sports day has 3 stickers plus x more; altogether there are 29. What is x?**

A) 28 B) 26 C) 29 D) 3

7 **A box in a craft workshop has 14 stickers plus x more; altogether there are 36. What is x?**

A) 14 B) 36 C) 24 D) 22

8 **A box at a school fair has 22 stickers plus x more; altogether there are 32. What is x?**

A) 22 B) 32 C) 12 D) 10

9 **A box in the class garden has 9 stickers plus x more; altogether there are 22. What is x?**

A) 9 B) 15 C) 22 D) 13

10 **A box at a market stall has 14 stickers plus x more; altogether there are 31. What is x?**

A) 17 B) 14 C) 31 D) 19

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 30	2. 28	3. 29
4. 7	5. 25	6. 16
7. 29	8. 24	9. 10
10. 23	11. 26	12. 22
13. 10	14. 13	15. 17

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Creating stories for equations

LEARNING GOAL Creating stories for equations

MATHS MODEL • EQUATIONS



Describe the rule, then explain why it works.

LEARN THE IDEA

An equation says two quantities are equal. use inverse operations to solve for the unknown.

WORKED EXAMPLE 1

Solve x + 14 = 26.

Answer: 12

x = 26 − 14 = 12.

WORKED EXAMPLE 2

Solve x + 20 = 46.

Answer: 26

x = 46 − 20 = 26.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Creating stories for equations

PLAY · LEARN THE STEPS

1. **Solve x + 17 = 41.**
   A) 41 B) 17 C) 26 D) 24
2. **Solve x + 10 = 37.**
   A) 27 B) 29 C) 10 D) 37
3. **Solve x + 9 = 37.**
   A) 9 B) 30 C) 28 D) 37
4. **Solve x + 13 = 39.**
   A) 28 B) 26 C) 39 D) 13
5. **Solve x + 4 = 29.**
   A) 25 B) 29 C) 27 D) 4

CHALLENGE · DIFFERENT EXAMPLES

1. **A box at a science exhibition has 8 stickers plus x more; altogether there are 18. What is x?**
   A) 8 B) 10 C) 12 D) 18
2. **A box during a reading competition has 5 stickers plus x more; altogether there are 18. What is x?**
   A) 18 B) 15 C) 13 D) 5
3. **A box for a school fundraiser has 17 stickers plus x more; altogether there are 28. What is x?**
   A) 11 B) 13 C) 28 D) 17
4. **A box at a farm has 8 stickers plus x more; altogether there are 18. What is x?**
   A) 18 B) 12 C) 8 D) 10
5. **A box during a sports day has 20 stickers plus x more; altogether there are 43. What is x?**
   A) 43 B) 23 C) 20 D) 25

Creating stories for equations

CHALLENGE · APPLY WHAT YOU KNOW

6. **A box in a craft workshop has 14 stickers plus x more; altogether there are 36. What is x?**
   A) 22 B) 36 C) 14 D) 24

7. **A box at a school fair has 22 stickers plus x more; altogether there are 47. What is x?**
   A) 47 B) 22 C) 27 D) 25

8. **A box in the class garden has 7 stickers plus x more; altogether there are 15. What is x?**
   A) 15 B) 8 C) 10 D) 7

9. **A box at a market stall has 16 stickers plus x more; altogether there are 36. What is x?**
   A) 22 B) 36 C) 16 D) 20

10. **A box during a library project has 22 stickers plus x more; altogether there are 26. What is x?**
    A) 6 B) 22 C) 4 D) 26

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 24	2. 27	3. 28
4. 26	5. 25	6. 10
7. 13	8. 11	9. 10
10. 23	11. 22	12. 25
13. 8	14. 20	15. 4

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Properties of squares and rectangles

LEARNING GOAL Properties of squares and rectangles

MATHS MODEL • QUADRILATERALS



LEARN THE IDEA

Rectangles have four right angles and equal opposite sides. a square also has four equal sides.

WORKED EXAMPLE 1

How many right angles does a rectangle have?

Answer: 4

Every rectangle has four right angles.

WORKED EXAMPLE 2

In a 11 cm by 4 cm rectangle, which pairs of sides are equal?

Answer: opposite sides

The two long sides are equal and the two short sides are equal.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Properties of squares and rectangles

PLAY · LEARN THE STEPS

1. **How many right angles does a rectangle have?**
   A) 0 B) 1 C) 4 D) 2
2. **In a 14 cm by 14 cm rectangle, which pairs of sides are equal?**
   A) diagonals only B) opposite sides C) no sides D) adjacent sides only
3. **In a 13 cm by 12 cm rectangle, which pairs of sides are equal?**
   A) diagonals only B) opposite sides C) adjacent sides only D) no sides
4. **In a 14 cm by 11 cm rectangle, which pairs of sides are equal?**
   A) opposite sides B) no sides C) adjacent sides only D) diagonals only
5. **In a 7 cm by 6 cm rectangle, which pairs of sides are equal?**
   A) diagonals only B) adjacent sides only C) no sides D) opposite sides

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangle during a reading competition is 9 cm long and 15 cm wide. Which pairs of sides have equal lengths?**
   A) opposite sides B) no sides C) adjacent sides only D) diagonals only
2. **A craft frame for a school fundraiser is a rectangle measuring 7 cm by 10 cm. How many right angles does it have?**
   A) 4 B) 2 C) 0 D) 1
3. **A rectangle at a farm is 7 cm long and 3 cm wide. Which pairs of sides have equal lengths?**
   A) opposite sides B) adjacent sides only C) no sides D) diagonals only
4. **A craft frame during a sports day is a rectangle measuring 3 cm by 6 cm. How many right angles does it have?**
   A) 0 B) 1 C) 2 D) 4
5. **A rectangle in a craft workshop is 3 cm long and 16 cm wide. Which pairs of sides have equal lengths?**
   A) diagonals only B) no sides C) opposite sides D) adjacent sides only

Properties of squares and rectangles

CHALLENGE · APPLY WHAT YOU KNOW

6 **A craft frame at a school fair is a rectangle measuring 5 cm by 10 cm. How many right angles does it have?**

A) 4 B) 2 C) 0 D) 1

7 **A rectangle in the class garden is 5 cm long and 12 cm wide. Which pairs of sides have equal lengths?**

A) diagonals only B) no sides C) opposite sides D) adjacent sides only

8 **A craft frame at a market stall is a rectangle measuring 9 cm by 8 cm. How many right angles does it have?**

A) 0 B) 1 C) 2 D) 4

9 **A rectangle during a library project is 10 cm long and 9 cm wide. Which pairs of sides have equal lengths?**

A) opposite sides B) diagonals only C) adjacent sides only D) no sides

10 **A craft frame on a football field is a rectangle measuring 7 cm by 2 cm. How many right angles does it have?**

A) 0 B) 1 C) 2 D) 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4	2. opposite sides	3. opposite sides
4. opposite sides	5. opposite sides	6. opposite sides
7. 4	8. opposite sides	9. 4
10. opposite sides	11. 4	12. opposite sides
13. 4	14. opposite sides	15. 4

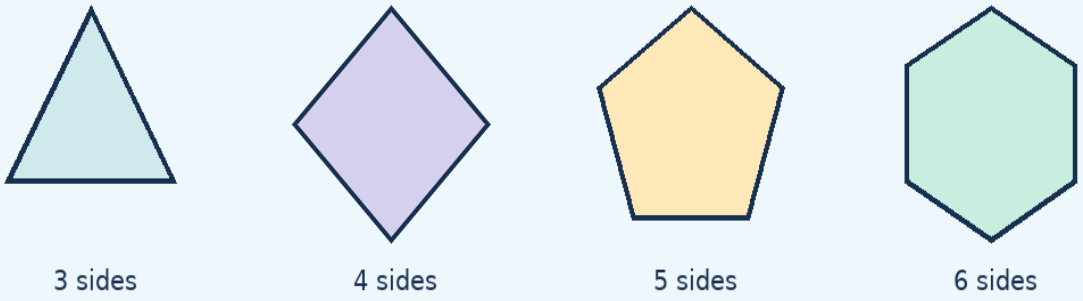
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Regular and irregular polygons

LEARNING GOAL Regular and irregular polygons

MATHS MODEL • POLYGON



LEARN THE IDEA

Regular polygons have every side equal and every interior angle equal; otherwise they are irregular.

WORKED EXAMPLE 1

Name a 5-sided polygon.

Answer: pentagon

A polygon with 5 sides has this name.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Regular and irregular polygons

PLAY · LEARN THE STEPS

1. **Name a 4-sided polygon.**
   A) triangle B) quadrilateral C) pentagon D) hexagon
2. **Name a 5-sided polygon.**
   A) hexagon B) triangle C) quadrilateral D) pentagon
3. **Name a 3-sided polygon.**
   A) quadrilateral B) hexagon C) triangle D) pentagon
4. **Name a 6-sided polygon.**
   A) hexagon B) pentagon C) triangle D) quadrilateral
5. **Name a 8-sided polygon.**
   A) octagon B) pentagon C) triangle D) quadrilateral

CHALLENGE · DIFFERENT EXAMPLES

1. **A regular polygon for a school fundraiser has 6 equal sides and angles. What is the name of its shape?**
   A) pentagon B) quadrilateral C) hexagon D) triangle
2. **A regular polygon at a farm has 3 equal sides and angles. What is the name of its shape?**
   A) hexagon B) pentagon C) quadrilateral D) triangle
3. **A quadrilateral during a sports day has all sides and all angles equal. Is it regular or irregular?**
   A) irregular B) regular C) circular D) not a polygon
4. **A regular polygon in a craft workshop has 4 equal sides and angles. What is the name of its shape?**
   A) hexagon B) triangle C) quadrilateral D) pentagon
5. **A regular polygon at a school fair has 3 equal sides and angles. What is the name of its shape?**
   A) quadrilateral B) triangle C) pentagon D) hexagon

Regular and irregular polygons

CHALLENGE · APPLY WHAT YOU KNOW

6 **A regular polygon in the class garden has 5 equal sides and angles. What is the name of its shape?**

A) triangle B) hexagon C) quadrilateral D) pentagon

7 **A hexagon at a market stall has all sides and all angles equal. Is it regular or irregular?**

A) not a polygon B) irregular C) circular D) regular

8 **A regular polygon during a library project has 3 equal sides and angles. What is the name of its shape?**

A) quadrilateral B) hexagon C) pentagon D) triangle

9 **A regular polygon on a football field has 8 equal sides and angles. What is the name of its shape?**

A) octagon B) pentagon C) triangle D) quadrilateral

10 **A quadrilateral in a recycling club has sides or angles of unequal sizes. Is it regular or irregular?**

A) not a polygon B) regular C) circular D) irregular

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. quadrilateral	2. pentagon	3. triangle
4. hexagon	5. octagon	6. hexagon
7. triangle	8. regular	9. quadrilateral
10. triangle	11. pentagon	12. regular
13. triangle	14. octagon	15. irregular

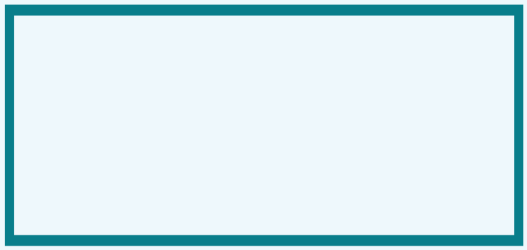
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Estimating and measuring perimeter

LEARNING GOAL Estimating and measuring perimeter

MATHS MODEL • PERIMETER



Walk around every edge

LEARN THE IDEA

Perimeter is the distance around a flat figure. add all sides or use 2(length + width).

WORKED EXAMPLE 1

Find the perimeter of a 4 cm by 10 cm rectangle.

Answer: 28 cm

P=2(l+w) gives 28 cm.

WORKED EXAMPLE 2

Find the perimeter of a 5 cm by 2 cm rectangle.

Answer: 14 cm

P=2(l+w) gives 14 cm.

TRY THIS IN REAL LIFE

Use paper squares or cubes to model dimensions; measure, calculate and explain your result.

Estimating and measuring perimeter

PLAY · LEARN THE STEPS

1. **Find the perimeter of a 8 cm by 7 cm rectangle.**
   A) 15 cm B) 30 cm C) 32 cm D) 56 cm
2. **Find the perimeter of a 13 cm by 2 cm rectangle.**
   A) 26 cm B) 32 cm C) 30 cm D) 15 cm
3. **Find the perimeter of a 20 cm by 14 cm rectangle.**
   A) 68 cm B) 280 cm C) 34 cm D) 70 cm
4. **Find the perimeter of a 18 cm by 6 cm rectangle.**
   A) 108 cm B) 50 cm C) 48 cm D) 24 cm
5. **Find the perimeter of a 13 cm by 10 cm rectangle.**
   A) 48 cm B) 130 cm C) 46 cm D) 23 cm

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangular garden at a farm measures 16 cm by 3 cm on a model. What is its perimeter?**
   A) 40 cm B) 19 cm C) 38 cm D) 48 cm
2. **A rectangular garden during a sports day measures 19 cm by 3 cm on a model. What is its perimeter?**
   A) 46 cm B) 57 cm C) 44 cm D) 22 cm
3. **A rectangular garden in a craft workshop measures 6 cm by 15 cm on a model. What is its perimeter?**
   A) 44 cm B) 42 cm C) 90 cm D) 21 cm
4. **A rectangular garden at a school fair measures 12 cm by 8 cm on a model. What is its perimeter?**
   A) 42 cm B) 20 cm C) 40 cm D) 96 cm
5. **A rectangular garden in the class garden measures 23 cm by 9 cm on a model. What is its perimeter?**
   A) 64 cm B) 66 cm C) 32 cm D) 207 cm

Estimating and measuring perimeter

CHALLENGE · APPLY WHAT YOU KNOW

6 **A rectangular garden at a market stall measures 6 cm by 8 cm on a model. What is its perimeter?**

A) 48 cm B) 28 cm C) 14 cm D) 30 cm

7 **A rectangular garden during a library project measures 15 cm by 2 cm on a model. What is its perimeter?**

A) 36 cm B) 30 cm C) 34 cm D) 17 cm

8 **A rectangular garden on a football field measures 21 cm by 14 cm on a model. What is its perimeter?**

A) 72 cm B) 70 cm C) 294 cm D) 35 cm

9 **A rectangular garden in a recycling club measures 23 cm by 3 cm on a model. What is its perimeter?**

A) 69 cm B) 54 cm C) 52 cm D) 26 cm

10 **A rectangular garden at a science exhibition measures 17 cm by 7 cm on a model. What is its perimeter?**

A) 48 cm B) 119 cm C) 50 cm D) 24 cm

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 30 cm	2. 30 cm	3. 68 cm
4. 48 cm	5. 46 cm	6. 38 cm
7. 44 cm	8. 42 cm	9. 40 cm
10. 64 cm	11. 28 cm	12. 34 cm
13. 70 cm	14. 52 cm	15. 48 cm

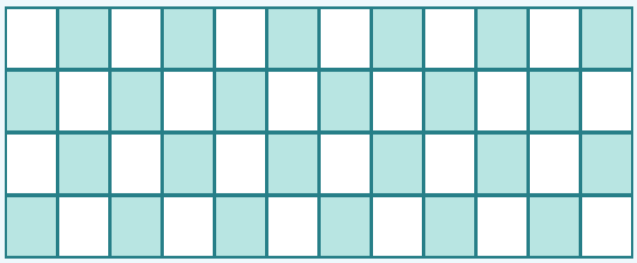
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding areas of rectangles

LEARNING GOAL Finding areas of rectangles

MATHS MODEL • AREA



12 columns × 4 rows = 48 square units

LEARN THE IDEA

Area is the surface covered in square units. for a rectangle, multiply length by width.

WORKED EXAMPLE 1

Find the area of a 5 cm by 6 cm rectangle.

Answer: 30 cm²

A=l×w gives 30 cm².

WORKED EXAMPLE 2

Find the area of a 18 cm by 12 cm rectangle.

Answer: 216 cm²

A=l×w gives 216 cm².

TRY THIS IN REAL LIFE

Use paper squares or cubes to model dimensions; measure, calculate and explain your result.

Finding areas of rectangles

PLAY · LEARN THE STEPS

1. **Find the area of a 11 cm by 9 cm rectangle.**
   A) 101 cm² B) 99 cm² C) 40 cm² D) 20 cm²
2. **Find the area of a 14 cm by 10 cm rectangle.**
   A) 24 cm² B) 140 cm² C) 48 cm² D) 142 cm²
3. **Find the area of a 13 cm by 4 cm rectangle.**
   A) 34 cm² B) 52 cm² C) 17 cm² D) 54 cm²
4. **Find the area of a 23 cm by 7 cm rectangle.**
   A) 163 cm² B) 161 cm² C) 30 cm² D) 60 cm²
5. **Find the area of a 11 cm by 15 cm rectangle.**
   A) 52 cm² B) 26 cm² C) 165 cm² D) 167 cm²

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangular garden during a sports day measures 14 cm by 6 cm on a model. What is its area?**
   A) 40 cm² B) 84 cm² C) 20 cm² D) 86 cm²
2. **A rectangular garden in a craft workshop measures 14 cm by 7 cm on a model. What is its area?**
   A) 42 cm² B) 100 cm² C) 98 cm² D) 21 cm²
3. **A rectangular garden at a school fair measures 23 cm by 11 cm on a model. What is its area?**
   A) 253 cm² B) 68 cm² C) 34 cm² D) 255 cm²
4. **A rectangular garden in the class garden measures 15 cm by 16 cm on a model. What is its area?**
   A) 242 cm² B) 31 cm² C) 240 cm² D) 62 cm²
5. **A rectangular garden at a market stall measures 19 cm by 3 cm on a model. What is its area?**
   A) 57 cm² B) 22 cm² C) 44 cm² D) 59 cm²

Finding areas of rectangles

CHALLENGE · APPLY WHAT YOU KNOW

6 **A rectangular garden during a library project measures 12 cm by 7 cm on a model. What is its area?**

A) 38 cm² B) 84 cm² C) 86 cm² D) 19 cm²

7 **A rectangular garden on a football field measures 19 cm by 7 cm on a model. What is its area?**

A) 135 cm² B) 26 cm² C) 52 cm² D) 133 cm²

8 **A rectangular garden in a recycling club measures 5 cm by 13 cm on a model. What is its area?**

A) 36 cm² B) 18 cm² C) 65 cm² D) 67 cm²

9 **A rectangular garden at a science exhibition measures 16 cm by 13 cm on a model. What is its area?**

A) 210 cm² B) 29 cm² C) 208 cm² D) 58 cm²

10 **A rectangular garden during a reading competition measures 11 cm by 8 cm on a model. What is its area?**

A) 19 cm² B) 90 cm² C) 88 cm² D) 38 cm²

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 99 cm²	2. 140 cm²	3. 52 cm²
4. 161 cm²	5. 165 cm²	6. 84 cm²
7. 98 cm²	8. 253 cm²	9. 240 cm²
10. 57 cm²	11. 84 cm²	12. 133 cm²
13. 65 cm²	14. 208 cm²	15. 88 cm²

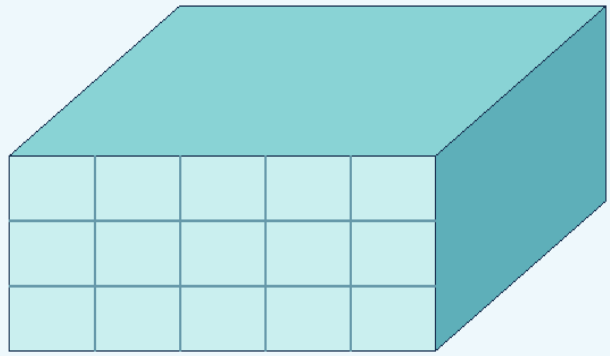
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Volume units and cubic blocks

LEARNING GOAL Volume units and cubic blocks

MATHS MODEL • VOLUME



Length × Width × Height = Volume

LEARN THE IDEA

Volume tells how many cubic units fit in a solid. multiply length × width × height.

WORKED EXAMPLE 1

Find the volume of a 5 × 4 × 2 cuboid.

Answer: 40 cm³

Volume = length × width × height = 5 × 4 × 2 = 40 cm³.

WORKED EXAMPLE 2

Find the volume of a 4 × 4 × 4 cuboid.

Answer: 64 cm³

Volume = length × width × height = 4 × 4 × 4 = 64 cm³.

TRY THIS IN REAL LIFE

Use paper squares or cubes to model dimensions; measure, calculate and explain your result.

Volume units and cubic blocks

PLAY · LEARN THE STEPS

1. **Find the volume of a 3 × 7 × 5 cuboid.**
   A) 21 cm³ B) 105 cm³ C) 15 cm³ D) 110 cm³
2. **Find the volume of a 8 × 7 × 4 cuboid.**
   A) 19 cm³ B) 224 cm³ C) 56 cm³ D) 228 cm³
3. **Find the volume of a 5 × 2 × 2 cuboid.**
   A) 9 cm³ B) 10 cm³ C) 22 cm³ D) 20 cm³
4. **Find the volume of a 5 × 7 × 3 cuboid.**
   A) 15 cm³ B) 105 cm³ C) 108 cm³ D) 35 cm³
5. **Find the volume of a 2 × 7 × 3 cuboid.**
   A) 42 cm³ B) 14 cm³ C) 12 cm³ D) 45 cm³

CHALLENGE · DIFFERENT EXAMPLES

1. **A storage box in a craft workshop holds 3 by 6 by 2 unit cubes. What is its volume?**
   A) 38 cm³ B) 36 cm³ C) 11 cm³ D) 18 cm³
2. **A storage box at a school fair holds 3 by 2 by 2 unit cubes. What is its volume?**
   A) 12 cm³ B) 7 cm³ C) 14 cm³ D) 6 cm³
3. **A storage box in the class garden holds 8 by 3 by 2 unit cubes. What is its volume?**
   A) 24 cm³ B) 48 cm³ C) 13 cm³ D) 50 cm³
4. **A storage box at a market stall holds 4 by 6 by 4 unit cubes. What is its volume?**
   A) 14 cm³ B) 100 cm³ C) 96 cm³ D) 24 cm³
5. **A storage box during a library project holds 8 by 7 by 4 unit cubes. What is its volume?**
   A) 56 cm³ B) 228 cm³ C) 19 cm³ D) 224 cm³

Volume units and cubic blocks

CHALLENGE · APPLY WHAT YOU KNOW

6. **A storage box on a football field holds 5 by 7 by 4 unit cubes. What is its volume?**
   A) 35 cm³ B) 140 cm³ C) 144 cm³ D) 16 cm³

7. **A storage box in a recycling club holds 3 by 5 by 3 unit cubes. What is its volume?**
   A) 45 cm³ B) 48 cm³ C) 11 cm³ D) 15 cm³

8. **A storage box at a science exhibition holds 5 by 5 by 5 unit cubes. What is its volume?**
   A) 25 cm³ B) 15 cm³ C) 130 cm³ D) 125 cm³

9. **A storage box during a reading competition holds 6 by 7 by 5 unit cubes. What is its volume?**
   A) 18 cm³ B) 215 cm³ C) 42 cm³ D) 210 cm³

10. **A storage box for a school fundraiser holds 7 by 4 by 3 unit cubes. What is its volume?**
    A) 14 cm³ B) 87 cm³ C) 28 cm³ D) 84 cm³

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 105 cm³
2. 224 cm³
3. 20 cm³
4. 105 cm³
5. 42 cm³
6. 36 cm³
7. 12 cm³
8. 48 cm³
9. 96 cm³
10. 224 cm³
11. 140 cm³
12. 45 cm³
13. 125 cm³
14. 210 cm³
15. 84 cm³

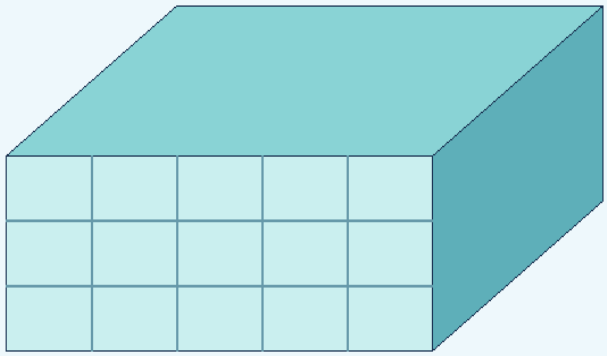
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Different boxes with the same volume

LEARNING GOAL Different boxes with the same volume

MATHS MODEL • BOX VOLUME



Length × Width × Height = Volume

LEARN THE IDEA

Different length-width-height triples can form boxes with the same volume when their products agree.

WORKED EXAMPLE 1

A cuboid has 24 unit cubes and a 4 × 2 base. Find its height.

Answer: 3

Equal volumes: 24 ÷ (4 × 2) = 3 layers.

WORKED EXAMPLE 2

A cuboid has 48 unit cubes and a 4 × 6 base. Find its height.

Answer: 2

Equal volumes: 48 ÷ (4 × 6) = 2 layers.

TRY THIS IN REAL LIFE

Use paper squares or cubes to model dimensions; measure, calculate and explain your result.

Different boxes with the same volume

PLAY · LEARN THE STEPS

1. **A cuboid has 84 unit cubes and a 3 × 7 base. Find its height.**
   A) 10 B) 5 C) 4 D) 21
2. **A cuboid has 125 unit cubes and a 5 × 5 base. Find its height.**
   A) 6 B) 5 C) 25 D) 10
3. **A cuboid has 240 unit cubes and a 8 × 6 base. Find its height.**
   A) 48 B) 14 C) 5 D) 6
4. **A cuboid has 50 unit cubes and a 5 × 5 base. Find its height.**
   A) 10 B) 25 C) 3 D) 2
5. **A cuboid has 72 unit cubes and a 3 × 6 base. Find its height.**
   A) 18 B) 5 C) 4 D) 9

CHALLENGE · DIFFERENT EXAMPLES

1. **A box at a school fair holds 128 cubes. Its base measures 8 cubes by 4 cubes. What height makes another box with the same volume?**
   A) 4 B) 12 C) 32 D) 5
2. **A box in the class garden holds 245 cubes. Its base measures 7 cubes by 7 cubes. What height makes another box with the same volume?**
   A) 14 B) 6 C) 5 D) 49
3. **A box at a market stall holds 28 cubes. Its base measures 7 cubes by 2 cubes. What height makes another box with the same volume?**
   A) 3 B) 14 C) 2 D) 9
4. **A box during a library project holds 24 cubes. Its base measures 2 cubes by 3 cubes. What height makes another box with the same volume?**
   A) 9 B) 6 C) 4 D) 5
5. **A box on a football field holds 54 cubes. Its base measures 6 cubes by 3 cubes. What height makes another box with the same volume?**
   A) 18 B) 9 C) 4 D) 3

Different boxes with the same volume

CHALLENGE · APPLY WHAT YOU KNOW

6 **A box in a recycling club holds 112 cubes. Its base measures 4 cubes by 7 cubes. What height makes another box with the same volume?**

A) 28 B) 4 C) 5 D) 11

7 **A box at a science exhibition holds 64 cubes. Its base measures 4 cubes by 4 cubes. What height makes another box with the same volume?**

A) 16 B) 5 C) 8 D) 4

8 **A box during a reading competition holds 108 cubes. Its base measures 6 cubes by 6 cubes. What height makes another box with the same volume?**

A) 36 B) 12 C) 4 D) 3

9 **A box for a school fundraiser holds 56 cubes. Its base measures 2 cubes by 7 cubes. What height makes another box with the same volume?**

A) 5 B) 14 C) 9 D) 4

10 **A box at a farm holds 50 cubes. Its base measures 2 cubes by 5 cubes. What height makes another box with the same volume?**

A) 10 B) 5 C) 7 D) 6

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4	2. 5	3. 5
4. 2	5. 4	6. 4
7. 5	8. 2	9. 4
10. 3	11. 4	12. 4
13. 3	14. 4	15. 5

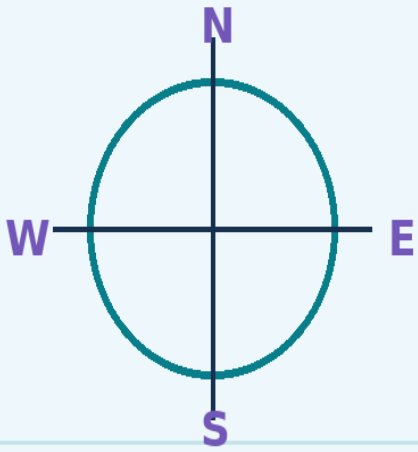
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing and measuring angles

LEARNING GOAL Comparing and measuring angles

MATHS MODEL • ANGLES



LEARN THE IDEA

Measure angles with a protractor: acute <90°, right =90°, obtuse between 90° and 180°, reflex >180°.

WORKED EXAMPLE 1

Classify an angle of 135°.

Answer: obtuse

135° is obtuse: acute <90°, right =90°, obtuse 90–180°, straight =180°, reflex >180°.

WORKED EXAMPLE 2

Classify an angle of 270°.

Answer: reflex

270° is reflex: acute <90°, right =90°, obtuse 90–180°, straight =180°, reflex >180°.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Comparing and measuring angles

PLAY · LEARN THE STEPS

1. **Classify an angle of 270°.**
   A) obtuse B) acute C) reflex D) right
2. **Classify an angle of 30°.**
   A) acute B) obtuse C) straight D) right
3. **Classify an angle of 150°.**
   A) obtuse B) right C) acute D) straight
4. **Classify an angle of 80°.**
   A) acute B) straight C) right D) obtuse
5. **Classify an angle of 110°.**
   A) straight B) acute C) obtuse D) right

CHALLENGE · DIFFERENT EXAMPLES

1. **A protractor reading in the class garden shows 180°. What type of angle is it?**
   A) obtuse B) straight C) acute D) right
2. **A protractor reading at a market stall shows 240°. What type of angle is it?**
   A) obtuse B) acute C) right D) reflex
3. **A protractor reading during a library project shows 30°. What type of angle is it?**
   A) straight B) obtuse C) right D) acute
4. **A protractor reading on a football field shows 240°. What type of angle is it?**
   A) acute B) right C) reflex D) obtuse
5. **A protractor reading in a recycling club shows 40°. What type of angle is it?**
   A) right B) obtuse C) straight D) acute

Comparing and measuring angles

CHALLENGE · APPLY WHAT YOU KNOW

6. **A protractor reading at a science exhibition shows 240°. What type of angle is it?**
   A) acute B) right C) obtuse D) reflex

7. **A protractor reading during a reading competition shows 90°. What type of angle is it?**
   A) straight B) acute C) right D) obtuse

8. **A protractor reading for a school fundraiser shows 150°. What type of angle is it?**
   A) right B) straight C) acute D) obtuse

9. **A protractor reading at a farm shows 270°. What type of angle is it?**
   A) reflex B) right C) acute D) obtuse

10. **A protractor reading during a sports day shows 90°. What type of angle is it?**
    A) straight B) right C) obtuse D) acute

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. reflex	2. acute	3. obtuse
4. acute	5. obtuse	6. straight
7. reflex	8. acute	9. reflex
10. acute	11. reflex	12. right
13. obtuse	14. reflex	15. right

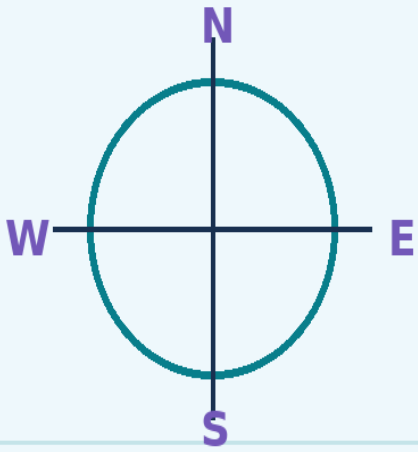
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

The four compass directions

LEARNING GOAL The four compass directions

MATHS MODEL • DIRECTION



LEARN THE IDEA

The four cardinal directions are north, east, south and west; a quarter-turn moves to the next direction.

WORKED EXAMPLE 1

Face north. Turn a quarter-turn left. Which way now?

Answer: west

Use the compass order N–E–S–W; after turning, face west.

WORKED EXAMPLE 2

Face west. Turn half a turn. Which way now?

Answer: east

Use the compass order N–E–S–W; after turning, face east.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

The four compass directions

PLAY · LEARN THE STEPS

1. **Face south. Turn a quarter-turn right. Which way now?**
   A) south B) north C) west D) east
2. **Face west. Turn half a turn. Which way now?**
   A) south B) west C) east D) north
3. **Face east. Turn a quarter-turn left. Which way now?**
   A) west B) east C) south D) north
4. **Face west. Turn a quarter-turn left. Which way now?**
   A) east B) west C) south D) north
5. **Face south. Turn a quarter-turn left. Which way now?**
   A) west B) north C) east D) south

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner faces south at a market stall and turns half a turn. Which direction is faced next?**
   A) south B) north C) west D) east
2. **A learner faces east during a library project and turns one quarter-turn to the left. Which direction is faced next?**
   A) west B) east C) north D) south
3. **A learner faces north on a football field and turns one quarter-turn to the right. Which direction is faced next?**
   A) east B) south C) west D) north
4. **A learner faces west in a recycling club and turns one quarter-turn to the left. Which direction is faced next?**
   A) south B) north C) east D) west
5. **A learner faces north at a science exhibition and turns one quarter-turn to the left. Which direction is faced next?**
   A) south B) west C) north D) east

The four compass directions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A learner faces north during a reading competition and turns half a turn. Which direction is faced next?**

A) east B) north C) west D) south

7 **A learner faces south for a school fundraiser and turns half a turn. Which direction is faced next?**

A) east B) south C) north D) west

8 **A learner faces south at a farm and turns half a turn. Which direction is faced next?**

A) south B) east C) north D) west

9 **A learner faces east during a sports day and turns one quarter-turn to the left. Which direction is faced next?**

A) west B) north C) south D) east

10 **A learner faces south in a craft workshop and turns half a turn. Which direction is faced next?**

A) east B) south C) west D) north

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. west	2. east	3. north
4. south	5. east	6. north
7. north	8. east	9. south
10. west	11. south	12. north
13. north	14. north	15. north

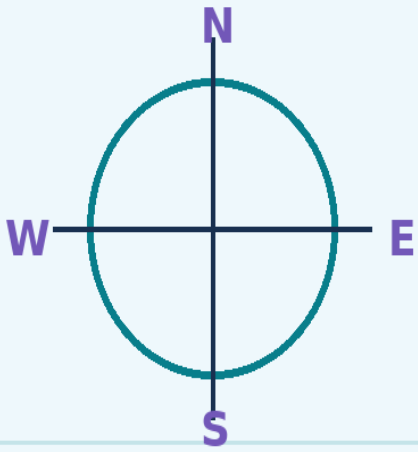
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Reflections and mirror images

LEARNING GOAL Reflections and mirror images

MATHS MODEL • REFLECTION



LEARN THE IDEA

A reflection is a mirror image across a line. every point stays the same distance from the line.

WORKED EXAMPLE 1

Reflect (2, 6) in the y-axis.

Answer: (-2, 6)

Reflection in the y-axis changes the sign of x only.

WORKED EXAMPLE 2

Reflect (4, 2) in the y-axis.

Answer: (-4, 2)

Reflection in the y-axis changes the sign of x only.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Reflections and mirror images

PLAY · LEARN THE STEPS

1. **Reflect (8, 3) in the y-axis.**
   A) (8, -3) B) (-8, 3) C) (-8, -3) D) (3, -8)
2. **Reflect (8, 7) in the y-axis.**
   A) (8, -7) B) (-8, -7) C) (-8, 7) D) (7, -8)
3. **Reflect (4, 5) in the y-axis.**
   A) (5, -4) B) (-4, 5) C) (4, -5) D) (-4, -5)
4. **Reflect (5, 2) in the y-axis.**
   A) (-5, -2) B) (5, -2) C) (-5, 2) D) (2, -5)
5. **Reflect (1, 8) in the y-axis.**
   A) (-1, 8) B) (8, -1) C) (-1, -8) D) (1, -8)

CHALLENGE · DIFFERENT EXAMPLES

1. **A point during a library project has coordinates (7, 2). Reflect it across the vertical y-axis. What is the new point?**
   A) (7, -2) B) (-7, 2) C) (-7, -2) D) (2, -7)
2. **A point on a football field has coordinates (8, 2). Reflect it across the vertical y-axis. What is the new point?**
   A) (-8, -2) B) (2, -8) C) (-8, 2) D) (8, -2)
3. **A point in a recycling club has coordinates (7, 7). Reflect it across the vertical y-axis. What is the new point?**
   A) (-7, -7) B) none of these C) (7, -7) D) (-7, 7)
4. **A point at a science exhibition has coordinates (5, 3). Reflect it across the vertical y-axis. What is the new point?**
   A) (5, -3) B) (3, -5) C) (-5, 3) D) (-5, -3)
5. **A point during a reading competition has coordinates (2, 2). Reflect it across the vertical y-axis. What is the new point?**
   A) (-2, 2) B) none of these C) (-2, -2) D) (2, -2)

Reflections and mirror images

CHALLENGE · APPLY WHAT YOU KNOW

6 **A point for a school fundraiser has coordinates (6, 8). Reflect it across the vertical y-axis. What is the new point?**

A) (8, -6) B) (6, -8) C) (-6, -8) D) (-6, 8)

7 **A point at a farm has coordinates (6, 5). Reflect it across the vertical y-axis. What is the new point?**

A) (-6, -5) B) (5, -6) C) (6, -5) D) (-6, 5)

8 **A point during a sports day has coordinates (6, 3). Reflect it across the vertical y-axis. What is the new point?**

A) (-6, -3) B) (-6, 3) C) (6, -3) D) (3, -6)

9 **A point in a craft workshop has coordinates (2, 1). Reflect it across the vertical y-axis. What is the new point?**

A) (1, -2) B) (-2, 1) C) (2, -1) D) (-2, -1)

10 **A point at a school fair has coordinates (8, 4). Reflect it across the vertical y-axis. What is the new point?**

A) (-8, 4) B) (8, -4) C) (4, -8) D) (-8, -4)

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. (-8, 3)	2. (-8, 7)	3. (-4, 5)
4. (-5, 2)	5. (-1, 8)	6. (-7, 2)
7. (-8, 2)	8. (-7, 7)	9. (-5, 3)
10. (-2, 2)	11. (-6, 8)	12. (-6, 5)
13. (-6, 3)	14. (-2, 1)	15. (-8, 4)

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

First-hand and second-hand data

LEARNING GOAL First-hand and second-hand data

MATHS MODEL • DATA SOURCE



LEARN THE IDEA

First-hand data is collected directly. second-hand data comes from published or previously collected sources.

WORKED EXAMPLE 1

Is studying published census data collecting first-hand or second-hand data?

Answer: second-hand

Obtained from a published source means second-hand data.

WORKED EXAMPLE 2

Is reading a newspaper table collecting first-hand or second-hand data?

Answer: second-hand

Obtained from a published source means second-hand data.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

First-hand and second-hand data

PLAY · LEARN THE STEPS

1. **Is interviewing farmers directly collecting first-hand or second-hand data?**
   A) first-hand B) guesswork C) second-hand D) experimental error
2. **Is weighing produce in person collecting first-hand or second-hand data?**
   A) first-hand B) guesswork C) second-hand D) experimental error
3. **Is consulting a school annual report collecting first-hand or second-hand data?**
   A) first-hand B) guesswork C) experimental error D) second-hand
4. **Is reading research in a journal collecting first-hand or second-hand data?**
   A) first-hand B) second-hand C) experimental error D) guesswork
5. **Is measuring rainfall yourself collecting first-hand or second-hand data?**
   A) guesswork B) experimental error C) second-hand D) first-hand

CHALLENGE · DIFFERENT EXAMPLES

1. **A researcher on a football field collects data by studying published census data. Is this first-hand or second-hand data?**
   A) second-hand B) experimental error C) guesswork D) first-hand
2. **A researcher in a recycling club collects data by studying published census data. Is this first-hand or second-hand data?**
   A) second-hand B) guesswork C) experimental error D) first-hand
3. **A researcher at a science exhibition collects data by reading a newspaper table. Is this first-hand or second-hand data?**
   A) second-hand B) guesswork C) experimental error D) first-hand
4. **A researcher during a reading competition collects data by tallying vehicles at the gate. Is this first-hand or second-hand data?**
   A) guesswork B) first-hand C) second-hand D) experimental error
5. **A researcher for a school fundraiser collects data by surveying your class. Is this first-hand or second-hand data?**
   A) guesswork B) first-hand C) second-hand D) experimental error

First-hand and second-hand data

CHALLENGE · APPLY WHAT YOU KNOW

6 **A researcher at a farm collects data by counting the cars outside. Is this first-hand or second-hand data?**

A) guesswork B) second-hand C) experimental error D) first-hand

7 **A researcher during a sports day collects data by studying published census data. Is this first-hand or second-hand data?**

A) second-hand B) experimental error C) first-hand D) guesswork

8 **A researcher in a craft workshop collects data by studying published census data. Is this first-hand or second-hand data?**

A) second-hand B) first-hand C) experimental error D) guesswork

9 **A researcher at a school fair collects data by measuring rainfall yourself. Is this first-hand or second-hand data?**

A) experimental error B) second-hand C) guesswork D) first-hand

10 **A researcher in the class garden collects data by tallying vehicles at the gate. Is this first-hand or second-hand data?**

A) guesswork B) first-hand C) second-hand D) experimental error

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. first-hand	2. first-hand	3. second-hand
4. second-hand	5. first-hand	6. second-hand
7. second-hand	8. second-hand	9. first-hand
10. first-hand	11. first-hand	12. second-hand
13. second-hand	14. first-hand	15. first-hand

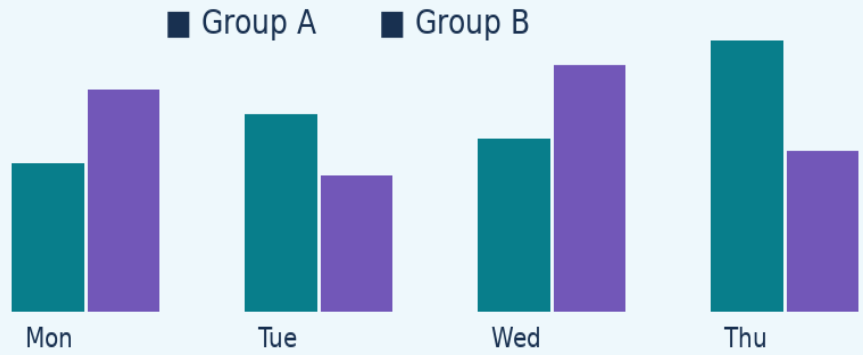
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Drawing and reading double-bar graphs

LEARNING GOAL Drawing and reading double-bar graphs

MATHS MODEL • DOUBLE BAR



LEARN THE IDEA

A double-bar graph places paired values side by side for comparison. always read both axes and the key.

WORKED EXAMPLE 1

One bar shows 30 and another 18. What is the difference?

Answer: 12

Compare the two bars: $|30-18|=12$.

WORKED EXAMPLE 2

One bar shows 23 and another 18. What is the difference?

Answer: 5

Compare the two bars: $|23-18|=5$.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Drawing and reading double-bar graphs

PLAY · LEARN THE STEPS

1. **One bar shows 30 and another 18. What is the difference?**
   A) 12 B) 18 C) 48 D) 14
2. **One bar shows 18 and another 26. What is the difference?**
   A) 10 B) 8 C) 44 D) 18
3. **One bar shows 32 and another 21. What is the difference?**
   A) 21 B) 13 C) 53 D) 11
4. **One bar shows 31 and another 32. What is the difference?**
   A) 1 B) 3 C) 63 D) 31
5. **One bar shows 37 and another 27. What is the difference?**
   A) 64 B) 12 C) 10 D) 27

CHALLENGE · DIFFERENT EXAMPLES

1. **Two bars in a recycling club show Class A = 29 and Class B = 10. Which class has the taller bar?**
   A) Class A B) cannot tell C) both equal D) Class B
2. **In a double-bar graph at a science exhibition, Class A recorded 14 books and Class B 11 books. What is the difference between their bars?**
   A) 11 B) 25 C) 5 D) 3
3. **In a paired bar chart during a reading competition, Class A collected 39 bottles and Class B collected 19. What is the combined total?**
   A) 58 B) 20 C) 63 D) 39
4. **In a double-bar graph for a school fundraiser, Class A recorded 36 books and Class B 36 books. What is the difference between their bars?**
   A) 2 B) 72 C) 0 D) 36
5. **In a double-bar graph at a farm, Class A recorded 20 books and Class B 24 books. What is the difference between their bars?**
   A) 6 B) 20 C) 44 D) 4

Drawing and reading double-bar graphs

CHALLENGE · APPLY WHAT YOU KNOW

6 **In a double-bar graph during a sports day, Class A recorded 33 books and Class B 31 books. What is the difference between their bars?**

A) 64 B) 4 C) 31 D) 2

7 **In a paired bar chart in a craft workshop, Class A collected 14 bottles and Class B collected 31. What is the combined total?**

A) 50 B) 17 C) 31 D) 45

8 **Two bars at a school fair show Class A = 33 and Class B = 30. Which class has the taller bar?**

A) cannot tell B) Class A C) both equal D) Class B

9 **In a double-bar graph in the class garden, Class A recorded 38 books and Class B 12 books. What is the difference between their bars?**

A) 28 B) 50 C) 12 D) 26

10 **In a paired bar chart at a market stall, Class A collected 10 bottles and Class B collected 37. What is the combined total?**

A) 52 B) 37 C) 27 D) 47

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 12	2. 8	3. 11
4. 1	5. 10	6. Class A
7. 3	8. 58	9. 0
10. 4	11. 2	12. 45
13. Class A	14. 26	15. 47

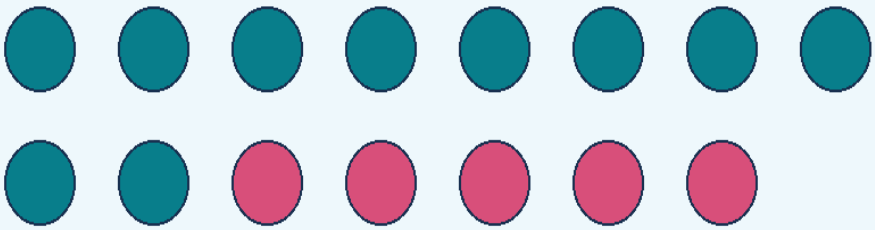
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Impossible, possible and certain

LEARNING GOAL Impossible, possible and certain

MATHS MODEL • CHANCE



Predict • Draw • Tally • Compare

LEARN THE IDEA

Impossible events cannot happen, certain events must happen, and possible events may happen.

WORKED EXAMPLE 1

Classify the event: rolling a 2 on an ordinary six-sided die.

Answer: possible

A standard die has faces 1–6; this outcome is possible.

WORKED EXAMPLE 2

Classify the event: rolling a 3 on an ordinary six-sided die.

Answer: possible

A standard die has faces 1–6; this outcome is possible.

TRY THIS IN REAL LIFE

Place safe coloured bottle tops in a bag, predict outcomes, then draw and tally twenty times.

Impossible, possible and certain

PLAY · LEARN THE STEPS

1. **Classify the event: rolling a zero on an ordinary six-sided die.**
   A) impossible B) not a mathematical likelihood C) certain D) possible
2. **Classify the event: rolling a 2 on an ordinary six-sided die.**
   A) certain B) impossible C) possible D) not a mathematical likelihood
3. **Classify the event: rolling a 3 on an ordinary six-sided die.**
   A) possible B) impossible C) certain D) not a mathematical likelihood
4. **Classify the event: rolling an eight on an ordinary six-sided die.**
   A) certain B) impossible C) not a mathematical likelihood D) possible
5. **Classify the event: rolling a 4 on an ordinary six-sided die.**
   A) impossible B) possible C) not a mathematical likelihood D) certain

CHALLENGE · DIFFERENT EXAMPLES

1. **A player at a science exhibition is rolling a zero on an ordinary six-sided die. How likely is this event?**
   A) not a mathematical likelihood B) impossible C) certain D) possible
2. **A player during a reading competition is rolling a 1 on an ordinary six-sided die. How likely is this event?**
   A) certain B) not a mathematical likelihood C) impossible D) possible
3. **A player for a school fundraiser is rolling a 3 on an ordinary six-sided die. How likely is this event?**
   A) impossible B) certain C) possible D) not a mathematical likelihood
4. **A player at a farm is rolling a zero on an ordinary six-sided die. How likely is this event?**
   A) impossible B) not a mathematical likelihood C) certain D) possible
5. **A player during a sports day is rolling a 3 on an ordinary six-sided die. How likely is this event?**
   A) certain B) impossible C) not a mathematical likelihood D) possible

Impossible, possible and certain

CHALLENGE · APPLY WHAT YOU KNOW

6 **A player in a craft workshop is rolling a zero on an ordinary six-sided die. How likely is this event?**

A) impossible B) possible C) not a mathematical likelihood D) certain

7 **A player at a school fair is rolling a 3 on an ordinary six-sided die. How likely is this event?**

A) impossible B) certain C) not a mathematical likelihood D) possible

8 **A player in the class garden is rolling a number from one to six on an ordinary six-sided die. How likely is this event?**

A) certain B) impossible C) not a mathematical likelihood D) possible

9 **A player at a market stall is rolling a 1 on an ordinary six-sided die. How likely is this event?**

A) impossible B) certain C) possible D) not a mathematical likelihood

10 **A player during a library project is rolling a 1 on an ordinary six-sided die. How likely is this event?**

A) not a mathematical likelihood B) impossible C) certain D) possible

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. impossible	2. possible	3. possible
4. impossible	5. possible	6. impossible
7. possible	8. possible	9. impossible
10. possible	11. impossible	12. possible
13. certain	14. possible	15. possible

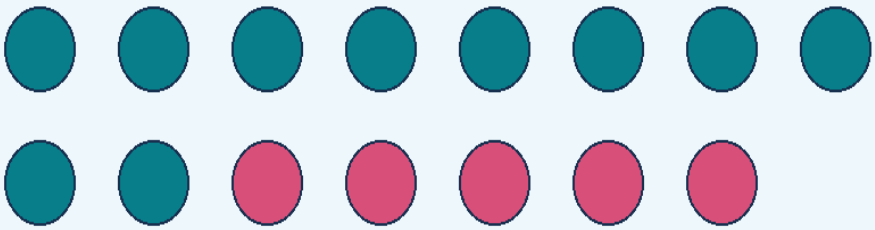
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Designing probability experiments

LEARNING GOAL Designing probability experiments

MATHS MODEL • PROBABILITY



Predict • Draw • Tally • Compare

LEARN THE IDEA

Predict and test an event using repeated fair trials. record all results and compare the observed frequencies.

WORKED EXAMPLE 1

Which is more likely: red (2) or blue (1) bottle tops?

Answer: red

Compare the counts; 2 red versus 1 blue.

WORKED EXAMPLE 2

Which is more likely: red (4) or blue (5) bottle tops?

Answer: blue

Compare the counts; 4 red versus 5 blue.

TRY THIS IN REAL LIFE

Place safe coloured bottle tops in a bag, predict outcomes, then draw and tally twenty times.

Designing probability experiments

PLAY · LEARN THE STEPS

1. **Which is more likely: red (10) or blue (6) bottle tops?**
   A) equally likely B) red C) blue D) neither can be picked
2. **Which is more likely: red (9) or blue (8) bottle tops?**
   A) neither can be picked B) equally likely C) blue D) red
3. **Which is more likely: red (8) or blue (1) bottle tops?**
   A) blue B) equally likely C) neither can be picked D) red
4. **Which is more likely: red (2) or blue (9) bottle tops?**
   A) blue B) equally likely C) neither can be picked D) red
5. **Which is more likely: red (10) or blue (5) bottle tops?**
   A) blue B) red C) equally likely D) neither can be picked

CHALLENGE · DIFFERENT EXAMPLES

1. **A bag during a reading competition has 8 red and 8 blue bottle tops. Which colour is more likely on one random draw?**
   A) equally likely B) neither can be picked C) red D) blue
2. **A bag for a school fundraiser has 1 red and 3 blue bottle tops. Which colour is more likely on one random draw?**
   A) neither can be picked B) red C) equally likely D) blue
3. **A bag at a farm has 1 red and 7 blue bottle tops. Which colour is more likely on one random draw?**
   A) blue B) neither can be picked C) equally likely D) red
4. **A bag during a sports day has 7 red and 1 blue bottle tops. Which colour is more likely on one random draw?**
   A) blue B) neither can be picked C) red D) equally likely
5. **A bag in a craft workshop has 6 red and 2 blue bottle tops. Which colour is more likely on one random draw?**
   A) equally likely B) neither can be picked C) red D) blue

Designing probability experiments

CHALLENGE · APPLY WHAT YOU KNOW

6 **A bag at a school fair has 1 red and 3 blue bottle tops. Which colour is more likely on one random draw?**

A) equally likely B) neither can be picked C) red D) blue

7 **A bag in the class garden has 1 red and 7 blue bottle tops. Which colour is more likely on one random draw?**

A) red B) equally likely C) neither can be picked D) blue

8 **A bag at a market stall has 10 red and 3 blue bottle tops. Which colour is more likely on one random draw?**

A) equally likely B) blue C) neither can be picked D) red

9 **A bag during a library project has 6 red and 3 blue bottle tops. Which colour is more likely on one random draw?**

A) equally likely B) neither can be picked C) red D) blue

10 **A bag on a football field has 10 red and 5 blue bottle tops. Which colour is more likely on one random draw?**

A) neither can be picked B) red C) equally likely D) blue

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. red	2. red	3. red
4. blue	5. red	6. equally likely
7. blue	8. blue	9. red
10. red	11. blue	12. blue
13. red	14. red	15. red

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.